

Reverberant Elastography for Developmental Biology

by
Christian Zevallos Delgado

A Dissertation submitted to the Department of Biomedical Engineering,
Cullen College of Engineering
in partial fulfillment of the requirements for the degree of

Doctor of Philosophy

in Biomedical Engineering

Chair of Committee: Kirill V. Larin, Ph.D.

Committee Member: Tianfu Wu, Ph.D.

Committee Member: Sergey Shevkoplyas, Ph.D.

Committee Member: David Mayerich, Ph.D.

Committee Member: Fernando Zvietcovich, Ph.D.

University of Houston
May 2026

Copyright 2026, Christian Zevallos Delgado

DEDICATION

I dedicate all my work and efforts to ever blessed and most pure Theotokos. I also dedicate this work to my family, who have taught me how to overcome everyday challenges.

I want to take a moment to dedicate this work to all the amazing people who have played a role in my personal and professional journey. Whether it was a brief encounter or a valuable lesson, your support has been instrumental in helping me reach this goal.

ACKNOWLEDGMENTS

I would like to express my sincere gratitude to Dr. Kirill Larin for his guidance and for providing me with the incredible opportunity to achieve one of my most significant professional goals. I am thankful for his willingness to allow me the freedom to conduct experiments and explore new ideas, which has enabled me to push the boundaries of knowledge and address challenges in my research.

I would like to express my heartfelt gratitude to Dr. Salavat Aglyamov for his endless patience in discussing new research ideas and for his unwavering support for my work. I also want to thank Dr. Manmohan Singh for his assistance in developing my research projects and for his guidance throughout my PhD program. Additionally, I would like to extend special thanks to my committee members, Dr. Zvietcovich, Dr. Wu, Dr. Mayerich, and Dr. Shevkoplyas, for their invaluable support and guidance.

I want to take a moment to thank my amazing collaborators who were always there to lend a hand during my research: Dr. Lekven, Dr. Scarcelli, and Dr. Manns. I truly appreciate your support!

I want to take a moment to express my heartfelt thanks to all the wonderful current and former members of the Biomedical Optics Lab for making me feel so welcome since I've joined. A special shout-out to Dr. Fernando Zvietcovich and Dr. Taye Mekonnen—I truly appreciate your mentorship, guidance, and support throughout my research and during my PhD journey.

ABSTRACT

Critical biomechanical processes shape the formation and maturation of embryos, altering their biomechanical properties and driving the architecture of organs. Studying structural modifications and changes in stiffness during embryo development is crucial for understanding various congenital diseases, such as anencephaly and spina bifida. Imaging methods such as optical coherence tomography (OCT), a high-resolution imaging technique, were used to capture the structural features of embryos. Additionally, the elastography extension of OCT, known as optical coherence elastography (OCE), is used to assess the biomechanical properties of the embryos throughout their development. In OCE, a technique utilizing a reverberant wave field, known as reverberant OCE (RevOCE), offers enhanced penetration capability and enables the assessment of the mechanical properties of highly heterogeneous tissues, such as embryos. The main aim of this dissertation was to develop RevOCE methods to assess the biomechanical properties of mouse and zebrafish embryos during their development. A noncontact method for performing RevOCE was developed using a design featuring multiple air-coupled ultrasound transducers, capable of producing a robust reverberant field, tested in gelatin and ex vivo porcine corneas. In addition, the

lateral and axial elastography resolutions of traditional wave-based OCE were compared with reverberant OCE on gelatin phantoms, where RevOCE showed a finer elastography resolution (axially and laterally) in comparison with the traditional wave-based OCE. Two common animal models were used to study the biomechanical properties of embryos during their development. First, mouse embryos were used to study the differences among *ex vivo* embryos at different stages during organogenesis (gestational day 8.5 to 10.5). The results showed significant differences in the spine, heart, and brain stiffness using multifocal RevOCE. Furthermore, RevOCE was used to map the biomechanical properties of zebrafish embryos and larvae (0 hours post-fertilization to 5 days post-fertilization). In conclusion, during embryonic development, several dynamic biomechanical modifications occur in the forming tissue and organs. RevOCE is a robust technique capable of mapping these mechanical modifications completely non-invasively, providing a technique for a longitudinal assessment of embryo mechanical properties.

TABLE OF CONTENTS

DEDICATION	iii
ACKNOWLEDGMENTS	iv
ABSTRACT	v
LIST OF TABLES	ix
LIST OF FIGURES	x
CHAPTER 1 – INTRODUCTION	1
1.1 Developmental biology and biomechanical changes	1
1.1.1 Mouse embryo development.....	3
1.1.2 Zebrafish embryo and larvae development.....	5
1.2 Elastography.....	6
1.3 Optical coherence tomography (OCT).....	7
1.4 Optical coherence elastography (OCE).....	10
1.4.1 Wave based OCE and Reverberant OCE.....	12
1.5 Organization of this Dissertation	15
CHAPTER 2 – NONCONTACT AIR-COUPLED REVOCE	16
2.1 Introduction.....	16
2.2 Materials and Methods.....	17
2.3 Results.....	22
2.4 Discussion	27
2.5 Conclusions	28
CHAPTER 3 – ADVANTAGES OF REVOCE COMPARED WITH A CONVENTIONAL UNIDIRECTIONAL WAVE-BASED OCE.....	30
3.1 Introduction.....	30
3.2 Material and Methods	31

3.3 Results	33
3.4 Discussion	44
3.5 Conclusions	45
CHAPTER 4 – USE OF REVOCE TO MAP THE MECHANICAL PROPERTIES OF DEVELOPING MOUSE EMBRYOS	46
4.1 Introduction	46
4.2 Materials and Methods	48
4.3 Results	49
4.4 Discussion	53
4.5 Conclusions	54
CHAPTER 5 – USE OF REVOCE TO ASSESS THE BIOMECHANICAL PROPERTIES OF EMBRYO AND LARVAL ZEBRAFISH	55
5.1 Introduction	55
5.2 Materials and Methods	56
5.3 Results	58
5.4 Discussion	61
5.5 Conclusions	62
CHAPTER 6 – CONCLUSIONS	63
REFERENCES	66

LIST OF TABLES

Table 3-1. Elastography resolution for traditional wave-based elastography across all phantom arrangements.....	38
Table 3.2. Elastography resolution for RevOCE elastography across all phantom configurations.....	43

LIST OF FIGURES

Figure 2.1. Schematic of the setup used to measure the pressure field of the ACUS transducers.....	18
Figure 2.2. 3D representation of the ACUS array and 3D printed holding ring alongside an image of the device.....	19
Figure 2.3. PhS-SDOCT system setup used for RevOCE imaging. C: collimator, GS: 2D Galvo scanner, M: mirror, P: aperture, SL: scan lens. SLD: superluminescent diode.....	20
Figure 2.4. Normalized acoustic pressure distribution at the focal plane (FWHM = 4.37 mm) of a single transducer.....	23
Figure 2.5. Normalized acoustic pressure distribution at the focal xy-plane produced by the five-transducer array.....	23
Figure 2.6. RevOCE imaging of a tissue-mimicking gelatin phantom. a) 3D OCT image of a phantom with two lateral layers b) Snapshots of the reverberant shear field at the indicated time points. The color bar refers to the axial particle velocity.....	24
Figure 2.7. a) 3D-spatial wave speed map showing elastogram. A ROI was selected to produce the lateral shear wave speed profile. b) Lateral wave speed profile and sigmoid fit at the transition between soft and stiff region (n = 3 phantoms).....	25
Figure 2.8. 3D OCT structural volume and B-mode image of the porcine cornea. b 3D volume snapshots of the reverberant shear field at the indicated time point after excitation.....	26
Figure 2.9. 3D volume of the shear wave speed map of the cornea and 2D wave speed slice.....	26
Figure 3.1. a) Generation of shear waves. b) Generation of the reverberant shear field. c) Heterogeneous gelatin phantoms (4% and 8%, w/w) in different arrangements: i) side-by-side, ii) up-down, and iii) L-shaped.....	33
Figure 3.2. OCT images of the phantoms assessed at a) 1 kHz, d) 5 kHz, and g) 10 kHz. Wave propagation snapshots for b) 1 kHz, e) 5 kHz, and h) 10 kHz. Local wave speed maps at c) 1 kHz, f) 5 kHz, i) and 10 kHz. Scale bars are 1mm.	34
Figure 3.3. OCT images of the phantoms assessed at a) 1 kHz, d) 5 kHz, and g) 10 kHz. Wave propagation snapshots for b) 1 kHz,	

e) 5 kHz, and h) 10 kHz. Local wave speed maps at c) 1 kHz, f) 5 kHz, i) and 10 kHz. Scale bars are 1 mm..	35
Figure 3.4. OCT images of the phantoms assessed at a) 1 kHz, d) 5 kHz, and g) 10 kHz. Wave propagation snapshots for b) 1 kHz, e) 5 kHz, and h) 10 kHz. Local wave speed maps at c) 1 kHz, f) 5 kHz, i) and 10 kHz. Scale bars are 500 μ m.....	36
Figure 3.5. Distance of excitation sources for each frequency used in RevOCE.	38
Figure 3.6. a) Particle velocity maps snapshots for each frequency (1 kHz, 5 kHz, and 10 kHz) for each gelatin concentration (4% and 8%). b) Wave speed of a given correlation window size for each gelatin concentration and frequency.....	39
Figure 3.7. Images of side-by-side phantoms for a) 1 kHz, d) 5 kHz, and g) 10 kHz. RevOCE wave propagation snapshots for b) 1 kHz, e) 5 kHz, and h) 10 kHz. Local wave speed maps at c) 1 kHz, f) 5 kHz, i) and 10 kHz. Scale bar 1 mm.....	40
Figure 3.8. OCT images of up-down phantoms for a) 1 kHz, d) 5 kHz, and g) 10 kHz. RevOCE wave propagation for b) 1 kHz, e) 5 kHz, and h) 10 kHz. Local wave speed maps at c) 1 kHz, f) 5 kHz, i) and 10 kHz. Scale bar 1 mm.....	41
Figure 3.9. OCT images of L-shaped phantoms for a) 1 kHz, d) 5 kHz, and g) 10 kHz. RevOCE wave propagation for b) 1 kHz, e) 5 kHz, and h) 10 kHz. Local wave speed maps at c) 1 kHz, f) 5 kHz, i) and 10 kHz. Scale bar 1mm.....	42
Figure 4.1. Schematic for RevOCE on mouse embryos. The embryos were placed inside the embryonic culture media while the OCE scanning was executed.	49
Figure 4.2. Maximum intensity projections and 3D OCT images of mouse embryos at GD 8.5, 9.5, and 10.5, illustrating the main parts of the embryo, including brain formation, heart, and spine. Scale bar 500 μ m.	50
Figure 4.3. Snapshots depicting the normalized out-of-plane particle velocity at a centrally located sagittal slice for each of the embryos (GD 8.5, 9.5, and 10.5), illustrating the velocity maps of mouse embryos within the yolk sac. Scale bar 500 μ m.	51
Figure 4.4. Local shear wave speed maps of each of the embryos. The highlighted region by a black dashed region represents the brain in each mouse embryo developmental stage.....	52

Figure 4.5. Local wave speed average for each embryo of various developmental stages. The three regions (spine, heart, and brain) analyzed for each mouse embryo was analyzed in each case, and n=5 embryos per stage of development.	52
Figure 5.1. Schematic for RevOCE on zebrafish embryos. The embryos were placed on top of the agarose dish media while the OCE scanning was executed.	57
Figure 5.2. Maximum intensity projections of 3D OCT images of zebrafish embryos at 1 hpf, 24 hpf, 48 hpf, and 5 dpf. Scale bar 1 mm.	58
Figure 5.3. Snapshots depicting the normalized out-of-plane particle velocity at a centrally located sagittal slice for each of the embryos (1 hpf, 24 hpf, 48 hpf, and 5 dpf), illustrating the velocity maps of zebrafish embryos within the chorion. Scale bar 1 mm.	59
Figure 5.4. Local shear wave speed maps of each of the zebrafish embryos in each developmental stage (1 hpf, 24 hpf, 48 hpf, and 5 dpf).	60
Figure 5.5. Local shear wave speed average for each zebrafish embryo of various developmental stages. The whole-body speed was analyzed for each zebrafish embryo was analyzed in each case (n=3).	61

CHAPTER 1 – INTRODUCTION

This chapter examines the importance of assessing the biomechanical properties of tissues during embryonic development, with a particular focus on patterning. Moreover, this chapter will briefly discuss relevant anatomical modifications that tissue can undergo during its development, and introduce the technology implemented in this dissertation, optical coherence tomography (OCT) and its extension for mapping mechanical properties, optical coherence elastography (OCE).

1.1 Developmental biology and biomechanical changes

Various mechanical forces are instrumental in driving processes such as organogenesis [1, 2]. At the cellular scale, mechanical parameters such as cell adhesion and cellular viscoelastic properties can alter the shape of cells according to their types and aid in migration for tissue formation [3, 4]. These biomechanical forces driven in the first embryonic developmental stage constitute the emergence of the tissue mechanical environment [5].

During development, tissue architecture responds to both intrinsic and extrinsic forces that shape cellular growth. These forces can arise from membrane interactions, external deformation, osmotic pressure, and adhesion dynamics, following a spatiotemporally ordered interaction [6]. General developmental processes that undergo from a single cell to complex organisms, are influenced by the cell-cell interactions until complete organ maturation [7]. Forces such as tension or stretching allow cells to pull

neighbors and establish cell-cell junctions, and they drive tissue remodeling (tissue folding) [8]. Compression is a force that shapes cellular architecture, as some tissues are physically constrained in their growth, which regulates cellular proliferation [9]. On the other hand, compartment fluids interact with the cells and tissues from the early stages of development, carrying not only nutrients and waste, but also influencing the internal pressure of organisms, playing a dramatic role in organs and systems such as the cardiovascular system [10, 11]. In the late stages of development, while muscles start contracting, the skeleton and cartilage are forming [12].

Mice are primarily used to study development due to their high genetic (~85%) and physiological resemblance to humans [13]. It is an animal model that has a rapid gestational cycle (~ 20 days), high productivity, and is easy to manipulate and maintain [14]. On the other hand, other non-mammal animals can also be used to study development, such as zebrafish. In comparison to mouse models, zebrafish possess a ~70% genetic resemblance to humans and share around 84% of human disease-related genes [15, 16]. Zebrafish also have several other advantages as animal models in research. They have a rapid growth rate; fertilization occurs within 24 hours, and their external development is beneficial for genetic manipulation and imaging. Additionally, they reach the larval stage in just 5 days [17]. Furthermore, zebrafish are optically transparent in their initial days of development, which represents a higher advantage for easier manipulation and observation of their phenotypic characteristics [18, 19].

1.1.1 Mouse embryo development

The mouse gestation period is approximately 20 days, during which they progress through several stages where various tissues and organs form to create the final structure of a mouse [20]. It starts the pre-implantation period with fertilization after mating, where a sperm fertilizes an oocyte, forming a zygote [21]. In initial gestational days (GD), the zygote undergoes multiple and rapid mitotic divisions (~ 8 to 16 cells), forming the morula (GD 1.5 to 2.5), which later compacts, polarizes, and cavitates to form a blastocyst [22]. The blastocyst consists of a trophoblast layer and an inner cell mass during days 2.5 to 3.5 of gestation. Before the post-implantation process begins, a critical hatching process occurs, which is necessary for implantation. This process involves physical factors such as the rupture of the zona pellucida and the expansion of the blastocyst cavity. Hatching defects can result in embryo loss and infertility [23]. Once the blastocyst is fully formed, it is implanted in the uterine wall on GD 4.5 to 5, and after implantation, gastrulation begins, where the inner cell mass starts its differentiation to three different germ layers: endoderm, mesoderm, and ectoderm (GD 6 to 7.5) [24]. The newly formed cells differentiate from precursors of various organs and tissues in their respective germ layers [25, 26]. From GD 7.5 to ~ 13.5 , the rapid formation of organs is completed until the complete formation of the fetus and the end of the gestational period. During this process, the spatial arrangement of cells and differentiation into distinct tissues, involving multiple cellular proliferation and cell-cell interactions, occur until the full formation and maturation of organs [27]. During early organogenesis (GD 7.5 to 9.5), the formation of the embryo starts with the formation of

the neural plate and the head [28]. Also, the primordial cardiogenic plaque forms with the initial optic and gut formation [29]. From GD 9 to 11, in mid-organogenesis, the three primary brain vesicles are formed (forebrain, midbrain, and hindbrain) with the development of the liver, lungs, and the formation of the tail bud, limbs, and branchial arches [30]. In the late-organogenesis (GD 12 to 13.5), major events are complete, such as the formation of the brain sensory system and defined, and there is a clear differentiation of the stomach, liver, kidneys, lung, and pancreas [31]. In the late part of the gestational period, GD 14 to ~20, complete fetal maturation is completed, where the fetus grows rapidly in preparation for the birth [32].

From the early stages of pre-implantation, when only a few cells are present, to the maturation of the fetus and its eventual birth, various mechanical forces significantly influence the development of every tissue and organ in the mouse embryo. However, measuring these biomechanical properties presents a considerable challenge, and several techniques have been developed to study these properties in embryonic tissues. Common approaches involve using direct physical contact, such as atomic force microscopy (AFM) [33, 34], microindentation [35], and micropipette [36], and exotic methods such as laser ablation [37, 38]. While these techniques offer important insights into the mechanical properties of embryos during their development, alternative noncontact methods, like Brillouin microscopy, can be preferable since they can ensure the embryos' integrity throughout the developmental process [39-42]. Other non-destructive elastography techniques used to obtain the biomechanical properties of tissues are ultrasound elastography [43], light-sheet elastography [44], and OCE [45].

1.1.2 Zebrafish embryo and larvae development

Zebrafish develop rapidly; they transition from early stages of development (zygotic) to a completed form in the lapse of ~3 days post fertilization (dpf). The zygotic stage after fertilization, 0 to 0.75 hours post fertilization (hpf), is a single cell stage with a diameter of ~700 μm [46]. In the following hours (0.75 to 2.25 hpf), synchronous cell divisions occur, going through several phases (2, 4, 8, 16, 32, and 64 cells) every 15 minutes [47]. After cell division, the cellular reorganization gives rise to the appearance of the epiboly (2.25 to 5.25 hpf) in the blastula stage [48]. Around 5.25 to 10 hpf, in the gastrulation process, cells differentiate into three germ layers (ectoderm, mesoderm, and endoderm), forming the embryonic shield, where the cellular shape and behavior change according to their body plan [49]. The formation of the body starts around 10 to 24 hpf, the formation of the tail and the appearance of somites, and the organogenesis begins, a process called segmentation [50]. Different organs are derived from the 3 germ cell layers; neural cells are derived from the ectoderm, endothelial cells and muscle cells are derived from the mesoderm, and lung and intestinal epithelial cells are derived from the endoderm [51]. After 24 hpf (pharyngula stage), the heart starts beating, the head straightens, and pigmentation begins until 48 hpf, when the embryo finally breaks the chorion (hatching) and transitions to a free-swimming larva with a functional mouth [52]. At this stage, a fully functional zebrafish larva (72 hpf) starts seeking food, and the swim bladder inflates [53, 54].

While zebrafish develop, several complex biomechanical processes occur to complete the formation of the different growth stages [55]. Within these steps, cardiac

development is guided by mechanical tension in the heart tube [56, 57], and muscle development is signaled by somite formation for locomotion and zebrafish's posterior capacity to swim [58-60]. In addition, during development, the stiffness of the chorion (the membrane that surrounds the zebrafish while developing) reduces as the zebrafish grows and develops, probably due to the proteolytic activity of proteases released by the embryos [61]. Several methods are used to quantify the biomechanical properties of zebrafish during its development, such as mechanical indentation [62], Brillouin microscopy to study the zebrafish morphogenesis and spinal cord elongation [63, 64] and eye development [65], and optical coherence elastography [66].

1.2 Elastography

Elastography refers to biomedical imaging techniques that map and assess the stiffness of tissues [67, 68]. Initial elastography techniques relied on ultrasound images to track tissue motion [68, 69]. Later, images of tissue stiffness were created using a modified color Doppler technique, which was termed sonoelasticity. Although this was the first report of local stiffness mapping, its resolution was low [70, 71]. Over time, various imaging methods have been utilized in elastography, significantly enhancing the mechanical resolution of this technique across multiple fields. These imaging techniques include Brillouin microscopy [72, 73], ultrasound elastography (USE) [74, 75], magnetic resonance elastography (MRE) [76, 77], AFM [78], holographic imaging [79, 80], and OCE [81].

1.3 Optical coherence tomography

OCT is a non-invasive, high-resolution imaging modality that uses light to capture cross-sectional images. OCT was developed in the 1980s and was formalized in 1991 [82], and it is often called “optical ultrasound”, since it also relies on measuring the echo time delay of a backscattered signal to see inside samples [83]. However, due to the much higher speed of light and lack of techniques to directly measure echo time delays, OCT utilizes low coherence interferometry to create its hallmark detailed in-depth images [84, 85].

Most OCT systems consist of two main components: a sample arm and a reference arm. The sample arm directs a light beam into the sample and captures the backscattered light, which varies in delay based on the internal layers and structures of the sample. In contrast, the reference arm provides a known baseline using a mirror that reflects the light. When the light from both arms is recombined, it generates an interference pattern that is recorded by a detector. The signal captured at a single position is called an A-scan, while a series of consecutive A-scans taken laterally is referred to as a B-scan [86]. An important parameter in OCT is the resolution in depth, or axial resolution (Δz), which is inversely related to the light source bandwidth and is described as:

$$\Delta z = \left(\frac{2 \ln 2}{\pi} \right) \left(\frac{\lambda_0^2}{\Delta \lambda} \right), \quad (1.1)$$

where λ_0 and $\Delta \lambda$ are the central wavelength and the bandwidth of the light source, respectively for a source with a Gaussian spectrum [87]. Among the different kinds of

sources used in OCT are the super-luminescent diodes (SLD), a typical light source in OCT because of their unique properties, such as broadband light, low coherence, and high brightness [88, 89] and are common on spectral domain OCT. In swept-source optical coherence tomography, a high-speed tunable laser emitting narrow-band light is swept across a wide range of frequencies, enabling deep high-speed imaging [90].

OCT is typically divided into two main categories: time domain and Fourier domain [91, 92]. Time-domain OCT (TD-OCT) encompasses many of the first implementations of OCT, which were initially developed by moving the reference mirror to different positions, capturing one point at a time, either axially or laterally, in each step. However, TD-OCT has several disadvantages, including low-resolution images and relatively slow acquisition times (A-scan rates are limited to a few kHz), which made the method susceptible to motion artifacts [93-95]. In contrast, Fourier domain OCT (FD-OCT) can capture all depths simultaneously with a fixed mirror, resulting in significantly faster acquisition, improved resolution, and dramatically improved sensitivity and sensitivity roll-off [96-99].

In contrast to TD-OCT, which obtains the intensity of the entire interference pattern, FD-OCT captures the spectral interference, which is transformed by an inverse Fourier transform into axial scan information [100, 101]. FD-OCT is further divided into two methods: spectral domain OCT (SD-OCT) and swept-source OCT (SS-OCT) [102]. In SD-OCT, backscattered light passes through a diffraction grating before reaching a spectrometer, where a line-scan camera captures the entire spectrum [103]. The advantages of this modality include high-speed scanning for each A-scan (up to

~250 kHz with commercial cameras/spectrometers), with higher sensitivity and contrast than TD-OCT [104]. In SS-OCT, a tunable laser sweeps across a range of wavelengths, with a very narrow bandwidth for each wavelength. The spectral interference is usually detected with a high-speed balanced photodetector [105, 106]. SS-OCT often has even superior penetration than SD-OCT due to using longer wavelengths and almost no sensitivity roll-off with modern swept source lasers [107, 108]. In addition to its many biomedical applications, OCT has been utilized in a variety of non-medical applications, including material science [109], archeology [110], and art conservation [111]. In the medical field, OCT has been used primarily in ophthalmology [112], where it is now ubiquitous, but has also been utilized in cardiology [113], oncology [114], dermatology [115], neurology [116], and gastroenterology [117]. In the field of developmental biology, OCT has been widely used to characterize the development of embryos noninvasively [118, 119], with high resolution and speed, permitting the assessment of several major events such as neural tube closure in mouse embryos [120], embryo cardiac development [121, 122], and even imaging pre-implantation oocytes and embryos in the oviduct [123].

Beyond structural imaging, OCT has multiple extensions that greatly expand its utility, enabling high-resolution imaging of motion, stiffness, and tissue composition [124]. OCT can be used to perform angiography (OCT-A), where it is used to visualize the blood vessels completely label-free [125]. Polarization-sensitive OCT (PS-OCT) provides information about the tissue fibrillar organization, like collagen [126], and

spectroscopic OCT (SOCT) can analyze the full spectrum and glean tissue absorption properties potentially identifying oxygenation levels and cancerous changes [127].

1.4 Optical coherence elastography

OCE is the OCT extension that is used to measure and map the mechanical properties of tissues [81]. OCT measures the tissue deformation with high optical imaging resolution and sub-nanometer displacement sensitivity [128, 129]. In addition, OCE possesses high acquisition rates for 2D and 3D *in vivo* elastography and high imaging penetration [130-132].

OCE can be classified into two main groups based on how a mechanical load is applied to induce microscopic tissue movement, which is detected by OCT. This movement can then be mapped to determine the biomechanical properties of tissues using suitable mechanical models [133]. The methods can be categorized into static/quasistatic and dynamic approaches. In static/quasistatic OCE, measurements are primarily conducted using a uniaxial compression source. OCT images are collected before and after the compression process, allowing for the quantification of tissue displacement in these two states to calculate the axial strain [134, 135]. This method possesses a high mechanical contrast, and rigorous studies have shown that mechanical resolution is dependent on the mechanical loading and the deformation of the sample [136]. On the other hand, in dynamic OCE, tissue is rapidly vibrated either with a continuous or transient vibration, and the stimulation can be applied locally or across the whole sample. The most common type of dynamic OCE is wave-based OCE, where a localized mechanical wave is induced through the sample with a localized stimulation,

which is imaged by OCE, and the stiffness of tissues is estimated based on how fast this wave travels through the tissue. Several kinds of waves can travel in the sample at or beneath the surface, which can be divided into a few main categories. In soft biological tissues, body waves that can be purely longitudinal, also known as pressure waves (P-wave), have speeds predominantly in a range of 1400 to 1700 m/s [137], and the displacement is parallel to the direction of propagation. The speed of pressure waves is determined by the longitudinal elastic modulus, which is typically on the order of GPa in soft biological tissues. P-waves travel at speeds beyond OCT's capabilities, and therefore, are not usually considered in OCE. Transversely propagating waves, also known as shear waves, have displacements perpendicular to the wave propagation direction and have speeds from 0.1 to 50 m/s in soft tissues [138]. When shear waves propagate near the tissue surface, they are often referred to as guided waves, and could be further classified according to coupling medium and boundary conditions (e.g., Lamb waves [139], Rayleigh waves, and Scholte waves [140]). These are main types of waves imaged in OCE. The speed of shear waves is determined by the shear mechanical properties of the tissue, such as the shear elastic modulus, which is typically on the order of kilopascals (kPa) in soft biological tissues. This value is much smaller than the longitudinal elastic modulus due to the low compressibility of biological tissues.

In OCE, two primary acquisition protocols were developed to detect tissue motion and measure spatial motion distribution in both cross-sectional (2D) and volumetric (3D) formats over time [141, 142]. These protocols rely on motion acquisition, commonly referred to as M-mode. This involves the repetitive acquisition

of A-lines. The speed of acquisition is determined by the interval between two consecutive A-lines, known as time resolution. The acquisition speed with typical systems is up to a few hundred kHz, with some systems reaching superfast acquisition rates in the order of MHz [102, 143].

Two protocols are used for the A-line acquisition in OCE. The M-B mode acquisition protocol involves collecting multiple A-lines at a specific position, based on the desired spatial and temporal resolution, and synchronizing them with the excitation source. Once the A-lines for one position are acquired, the process moves to the next position, continuing until all desired lateral positions are covered. This results in a B-scan that includes multiple M-mode points, with each point containing a specified number of A-lines [144, 145]. In this method, the temporal resolution is the A-scan rate of the OCT system. The second acquisition protocol is called B-M mode, where A-lines are acquired laterally to build a single B-mode image, and multiple B-mode frames are acquired in the same position [146, 147]. In this paradigm, the A-scan rate is determined by the scan rate of the scanners in the system, and without ultra-fast systems, is usually orders of magnitude slower than M-B-mode imaging. Recently, B-M mode imaging has been used in the implementation of new methods in OCE, called asynchronous reverberant OCE (RevOCE) [148], enabling rapid RevOCE imaging.

1.4.1 Wave based OCE and reverberant OCE

In wave-based OCE, mechanical waves can be generated by various sources, which are classified into two categories: active and passive sources. Passive sources utilize natural, non-artificial means to assess the biomechanical properties of tissues.

This can be performed through methods such as respiration, cardiac pulse, and environmental noise. In contrast, active sources generate mechanical waves using an external source. Active methods can be further divided into contact and noncontact techniques, with noncontact methods being preferred for producing waves in delicate tissues, such as ocular tissues, and for *in vivo* applications [149-151]. On the other hand, contact methods depend on direct physical contact of sources with the sample to produce mechanical waves within the tissues [152]. Multidirectional methods such as standing waves, crawling waves [153], and reverberant fields [154], in addition to most unidirectional wave-based OCE methods, are also utilized to estimate the biomechanical properties of tissues.

In RevOCE, a reverberant field is created through the continuous interference of shear waves traveling in all directions. This method offers a higher axial mechanical resolution and mechanical contrast compared to other wave-based OCE methods [155]. This method was first reported in USE [154, 156-164] and later it was implemented in OCE [148, 155, 164-176] and MRE [164, 177-179]. In RevOCE, in a specific 2D z -plane, the particle velocity is used to calculate the local wave number (k_0) by:

$$R_{v_z v_z}(\Delta r_x) = C \left[j_0(k_0 \Delta r_x) - \frac{j_1(k_0 \Delta r_x)}{k_0 \Delta r_x} \right], \quad (1.2)$$

where j_0 and j_1 are the spherical Bessel functions of the 0th and 1st order, respectively, Δr_x is the autocorrelation direction along the x -axis, and C is a scalar amplitude constant. In relation to direction, Δr_x and Δr_y possess the same solution since both are

perpendicular to the measurement axis; however, in parallel case (Δr_z), the spatial autocorrelation is given by:

$$R_{v_z v_z}(\Delta r_z) = C \left[\frac{j_1(k_0 \Delta r_z)}{k_0 \Delta r_z} \right], \quad (1.3)$$

The local wavenumber (k_0) is then used to calculate the shear wave speed (c_s) by:

$$c_s = \frac{\omega_0}{k_0}, \quad (1.4)$$

where $\omega_0 = 2\pi f$, and f is the frequency of the wave. This method can be used to study shear wave speed in both directions in the xy plane, which is perpendicular to the excitation direction. However, when the measurement is taken parallel to the excitation (in the xz or yz planes), the shear wave speed is often underestimated compared to the results obtained when the measurement is made perpendicular to the source of excitation. [155, 180].

Several important parameters have been studied to understand the conditions required to induce an adequate reverberant field, such as the minimal excitation sources to achieve a complete reverberant field. A minimum of 60 incident plane waves is necessary to generate a reverberant shear wave field, thereby increasing the number of sources and waves that can potentially reduce the error in calculations [159, 173, 180]. Recently, new estimation methods have been developed to produce more accurate calculations of the shear wave speed using RevOCE, such as phase gradient [160], angular integration autocorrelation [164], the use of multifrequency shear waves [181], and the use asynchronous methodology integrated with phase gradient analysis, which

potentially reduces the time of acquisition and promotes the use of a single source and multiple sources [148, 171, 182].

1.5 Organization of this Dissertation

Chapter 2 details the implementation of completely noncontact air-coupled ultrasound RevOCE. Chapter 3 demonstrates the advantages of RevOCE compared with a conventional unidirectional wave-based OCE. Chapter 4 demonstrates the use of RevOCE to map the mechanical properties of developing mouse embryos, showing the development of mouse embryos while they remain in the yolk sac. Chapter 5 demonstrates the use of RevOCE to assess the biomechanical properties of embryo and larval zebrafish. Chapter 6 will describe the future applications of RevOCE, as well as methods for technical improvements for its extended applications.

CHAPTER 2 – NONCONTACT AIR-COUPLED REVOCE

In RevOCE, the interference of multiple shear waves simultaneously produced by several excitation sources generates a reverberant field within the sample. This chapter describes a completely noncontact RevOCE method using air-coupled ultrasound (ACUS) transducers. ACUS RevOCE was validated using gelatin phantoms and *ex vivo* porcine whole eye globes. The results of this *ex vivo* study suggest the viability of using ACUS transducers to produce completely noncontact RevOCE stimulation.

2.1 Introduction

In wave-based OCE, various contact and noncontact excitation sources produce waves on tissues [141]. In contrast to common wave-based OCE, where a single source produces a shear wave, RevOCE, an extension of wave-based OCE, has multiple waves traveling in all directions from a reverberant field to describe their biomechanical properties [183]. As mentioned in Chapter 1, reverberant elastography was first demonstrated using USE [154]. Later, using OCE, a reverberant field was induced in a sample tissue using a 3D printed ring with eight rods attached to a piezoelectric transducer bender [169, 175]. The rods slightly touched the sample, generating multiple shear waves in all directions; however, this contact-based arrangement can potentially produce damage to tissues or cause discomfort in sensitive tissues (e.g., *in vivo* cornea) [155]. As an alternative to the contact method, immersion-based multifocal acoustic radiation force (ARF) was employed to generate a reverberant field in the sample [172,

173]. The disadvantage of using indirect noncontact immersion ARF is that it needs a coupling medium (e.g., water or ultrasound gel) to permit the ultrasound waves to reach the sample [184]. On the other hand, ACUS is completely noncontact and uses air to transmit acoustic waves [149]. ACUS-OCE was extensively used to determine the biomechanical properties of sensitive tissue, such as ocular tissues [150, 185-190] and engineered skin scaffolds [191], demonstrating its capacity to produce guided waves in delicate specimens without causing significant damage. In addition to truly noncontact excitation, ACUS provides tight control over the spatiotemporal characteristics of stimulation, which is critical for robust mechanical quantitation [149, 192]. In this chapter, unfocused ACUS transducers are utilized to perform RevOCE. The main function of these inexpensive ACUS transducers is distance sensors [193]. Additionally, they were employed to create an acoustic resonator for measuring the index of refraction through defocusing shadowgraphy [194]. To demonstrate the feasibility of ACUS RevOCE, this method was tested on gelatin phantoms and *ex vivo* porcine corneas.

2.2 Materials and Methods

The unfocused transducers were characterized to understand their performance and be used in RevOCE. These transducers (TCT40-16, Shenzhen HiLetgo Technology Co., Ltd, China) possessed a nominal frequency of 40 kHz, a length of 12 mm, and a diameter of 16 mm [194]. The acoustic exposure levels of the transducers were characterized using a microphone (Eta 100 Ultra; XARION, Austria). The transducers were characterized using a driving signal of a single pulse with a duration of 0.5 ms modulating a continuous 40 kHz carrier signal generated with an arbitrary function

generator (DG4162, RIGOL Tech, China) and amplified by an RF power amplifier (1040L, Electronics & Innovation, Ltd., USA). As shown in Figure 2.1, A 3-axis motorized stage driven by a motion controller (ESP301, Newport Co., USA) was used to scan the xy-plane of maximum acoustic pressure to characterize the spatial properties of the acoustic pressure field of a single transducer and the arrangement of 5 transducers.

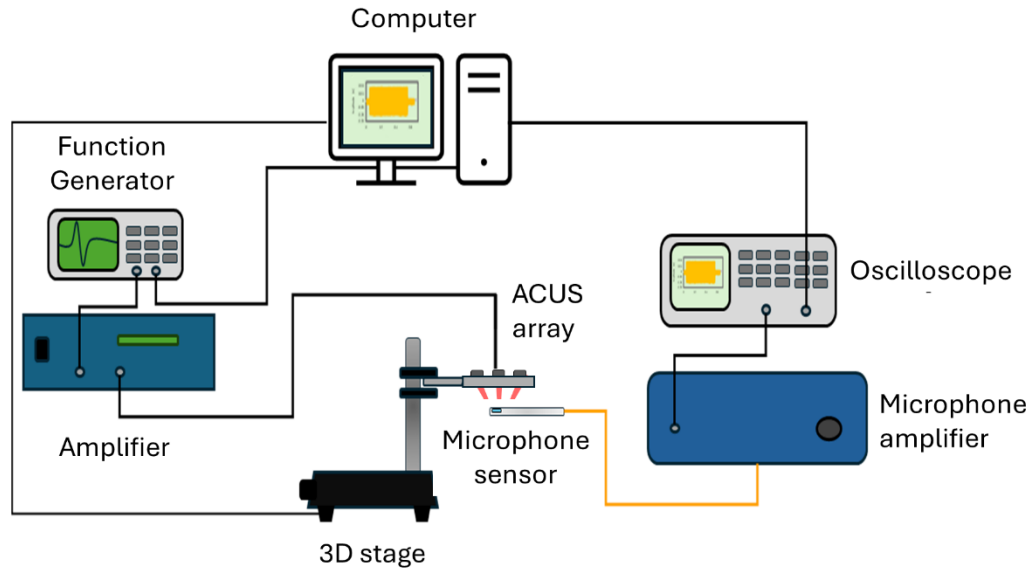


Figure 0.1. Schematic of the setup used to measure the pressure field of the ACUS transducers.

The acoustic pressure of a single transducer was characterized over a 10×10 mm² area at the transducer's maximum pressure plane. The detected pressure intensity was pre-amplified and transferred to a computer via a digital oscilloscope (DS4000, RIGOL Tech, China). A 3D printed ring of 4.5 cm external diameter and a thickness of 4 mm (Stratasys Objet Eden 260 v, Stratasys, Israel) was built to hold the array of five

transducers in a ring. The inner wall was beveled at a 45° inclination, and the five ACUS transducers were equally spaced in the ring (72° separation). All transducers were connected in a parallel configuration, and the acoustic pressure field was characterized as mentioned above, using a microphone over a $25 \times 25 \text{ mm}^2$ area at the transducer plane as determined by the location of maximum acoustic pressure (Figure 2.2).

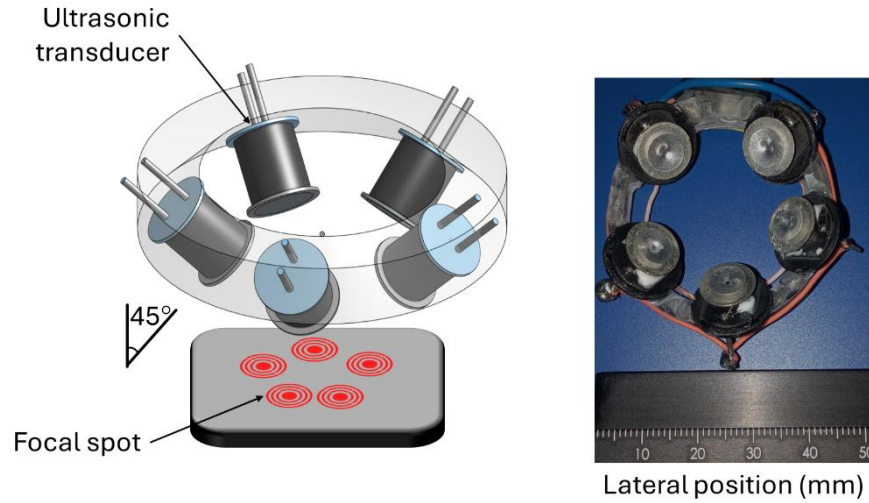


Figure 0.2. 3D representation of the ACUS array and 3D printed holding ring alongside an image of the device.

RevOCE imaging was performed with a phase-sensitive (PhS) SDOCT system. The system was based on a superluminescent diode (S48W0-B-I-20; Superlum Diodes Ltd., Ireland) with a central wavelength of 840 nm and a bandwidth of 50 nm, and with an axial and transverse resolution of $\sim 9 \text{ }\mu\text{m}$ and $\sim 8 \text{ }\mu\text{m}$ in air, respectively (Figure 2.3). The displacement sensitivity was 0.28 nm at an OCT signal-to-noise ratio $> 30 \text{ dB}$.

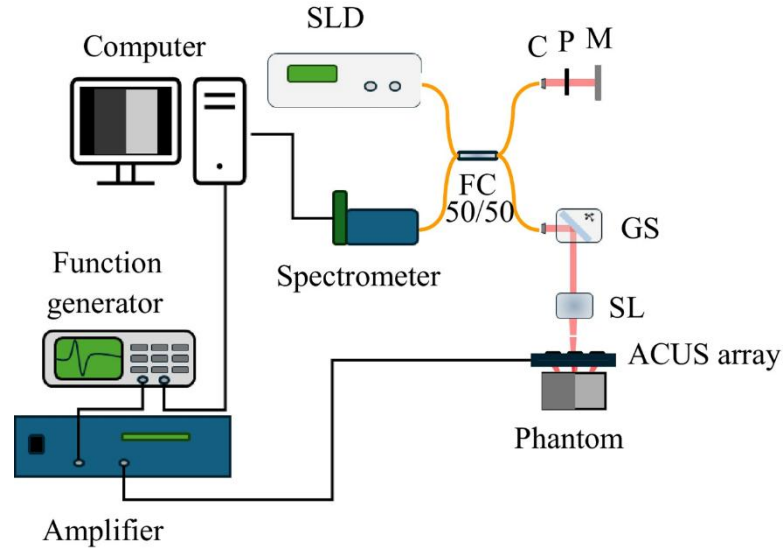


Figure 0.3. PhS-SDOCT system setup used for RevOCE imaging. C: collimator, GS: 2D Galvo scanner, M: mirror, P: aperture, SL: scan lens. SLD: superluminescent diode.

A side-by-side heterogeneous gelatin phantom with two distinct mechanical properties (4% and 8% gelatin concentrations) was used to test the capabilities of the multi-ACUS array to generate a reverberant field. The heterogeneous gelatin phantom was cast using porcine gelatin (Sigma-Aldrich, St. Louis, MO) mixed with TiO_2 (Pantai Chemical, GA) to improve scattering. Softer tissue was represented by the 4% w/w gelatin mixed with 0.05% w/w TiO_2 , and 8% w/w gelatin mixed with 0.1% w/w TiO_2 represented stiffer tissue. First, the 4% gelatin phantom was cast in a standard culture dish with a diameter of ~ 35 mm and a height of ~ 10 mm. Once solidified, half of the phantom was gently removed, and 8% w/w gelatin mix (almost at gelation

temperature) was poured into the empty space and cooled in a 4°C fridge to gel. Three such phantoms were cast.

A whole porcine cornea (less than 24 hours of enucleation) was obtained (Sioux-Preme Packing Co., Iowa City, IA) and mounted in a custom holder. The eye globe was cannulated for artificial intraocular pressure (IOP) control with a previously described system based on a closed loop feedback control of a micro-infusion pump and pressure transducer. The IOP was set to 10 mmHg during RevOCE imaging.

For the phantom experiments, RevOCE imaging was performed over an area of 9.2 mm and 12.6 mm in the x-axis and y-axis, respectively. A total of 500 A-lines per M-mode scan with 75×75 M-mode scan points were acquired in three-dimensional M-mode scanning [141]. In the case of the porcine cornea experiment, the scanning was performed over an area of 6.1 mm and 6.9 mm for the y-axis and x-axis, respectively. Here, 101×101 M-mode scan points were acquired with 500 A-lines per M-mode scan. Due to the size of the cornea and eye-globe, the excitation points were at or near the limbus [195], and the imaged region was at the apex of the cornea. In this experiment, the function generator (DG4162, Rigol Technologies, China) was used to generate a quasi-harmonic square pulse signal of ten cycles of 1 kHz (2 kHz for cornea) with a 50% duty cycle, which amplitude-modulated a continuous 40 kHz sinusoidal signal (i.e., the resonant frequency of the ACUS transducers). The modulated signal was amplified by an RF power amplifier (1040L, Electronics & Innovation, Ltd., USA) before driving the multi-ACUS array. The signal was synchronized to the OCT system frame trigger[144]. The A-scan rate was set to 50 kHz, so the temporal resolution was 20 μ s.

RevOCE data analysis was performed following the methods described by Zvietcovich *et al* [155], where the axial particle velocity was obtained from the axial temporal phase difference [196]. The surface of the gelatin phantom was tracked using intensity thresholding, and the motion under the surface was corrected for the refractive index mismatch [145]. The axial particle velocity in the gelatin phantoms was filtered at a frequency of 1 kHz, i.e., the excitation frequency, with a bandwidth of 200 Hz, using a first-order FIR filter with the aim of reducing irregularities during speed calculations [151]. A 2D spatial bandpass filter was utilized to minimize noise within the speed range of 0.2 m/s to 10.0 m/s in all particle velocity volumes. The 2D autocorrelation profiles were obtained over 1.75 mm² window size in each particle velocity volume, and the autocorrelation profiles were fitted to the analytical solution of the reverberant shear field [154, 156]. The same methodology was used in the cornea. After tracking the cornea surface and obtaining each in-depth layer, the particle velocity was filtered at a frequency of 2 kHz with a bandwidth of 200 Hz using a first-order FIR filter. The 2D autocorrelation for each cornea layer in the xy-plane was obtained using a window size of 1.5 mm² and fitted to the reverberant analytical solution (Eq. 1.2).

2.3 Results

The ACUS characterization results revealed that a single transducer, positioned 10 mm from the microphone sensor, produced a spot size of 4.37 mm (Figure 2.4).

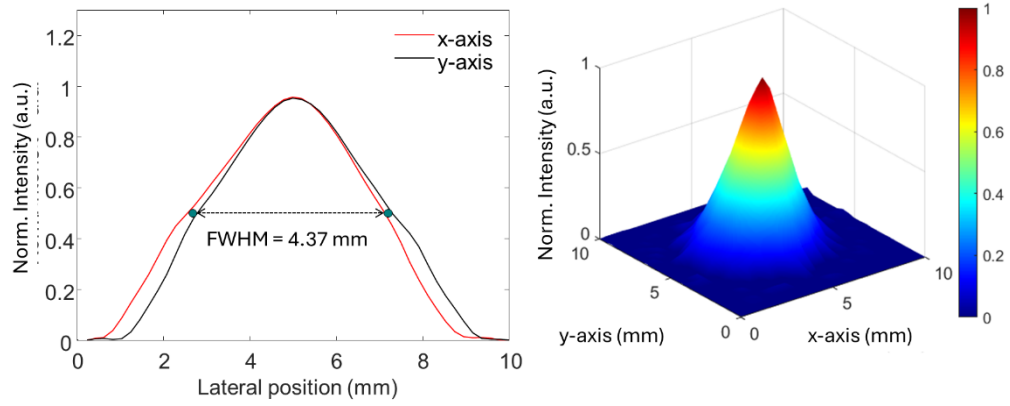


Figure 0.4. Normalized acoustic pressure distribution at the focal plane (FWHM = 4.37 mm) of a single transducer.

In the case of the multiACUS array, the five ACUS transducers exhibited a uniform acoustic pressure distribution, demonstrating the array's potential ability to create a reverberant field (Figure 2.5).

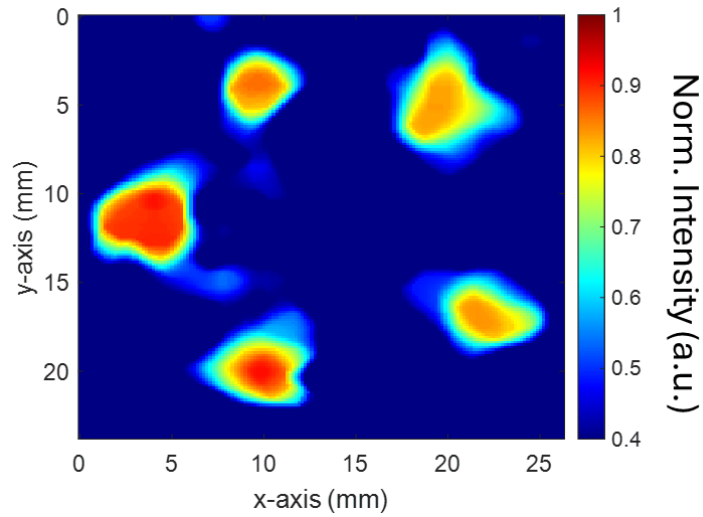


Figure 0.5. Normalized acoustic pressure distribution at the focal xy-plane produced by the five-transducer array.

In the case of gelatin phantoms (see Figure 2.6), the results showed that this method can create a robust reverberant field within the samples. As illustrated in Figure 2.6b, the motion volume snapshots reveal the reverberant wave field, emphasizing the distinct wavelengths between the softer and stiffer regions of the tissue-mimicking phantom. The soft region, consisting of 4% gelatin, exhibited an attenuated particle velocity compared to the stiffer region, which contained 8% gelatin. This difference can be attributed to the mechanical variations between the two sides of the phantom and the proximity of the excitation to the sample. This illustrates the effectiveness of the ACUS transducer array in generating a reverberant wave field.

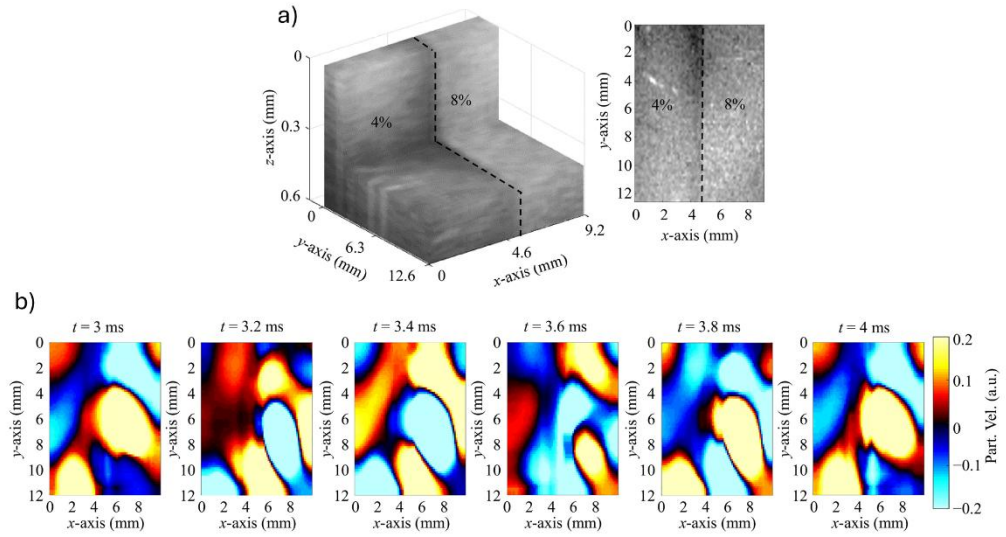


Figure 0.6. RevOCE imaging of a tissue-mimicking gelatin phantom. a) 3D OCT image of a phantom with two lateral layers b) Snapshots of the reverberant shear field at the indicated time points. The color bar refers to the axial particle velocity.

The axial particle velocity was utilized to calculate the wave speed in each xy-plane by fitting the 2D-autocorrelation profiles according to the standard RevOCE processing method[155]. The elastic contrast between the softer and stiffer regions in the phantoms is illustrated in Figure 2.7a, where the average elastic wave speed in the soft region (4% gelatin concentration) was 1.6 ± 0.1 m/s, and 2.6 ± 0.1 m/s in the stiff region (8% gelatin concentration). Figure 2.7b shows that the lateral elastic resolution was 208.8 ± 72.4 μm measured in three different side-to-side phantoms quantified as the full width at half maximum (FWHM) of a derivative of a sigmoid fitted to the speed profile [172] across the boxed area in Figure 2.7a.

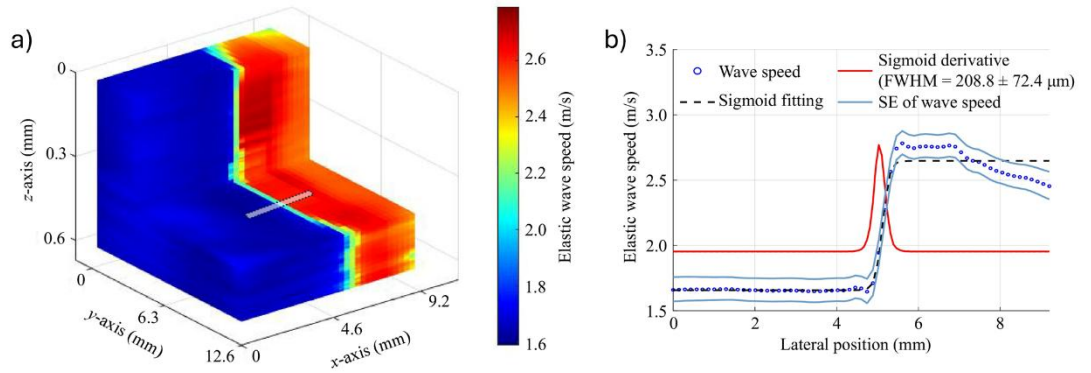


Figure 0.7. a) 3D-spatial wave speed map showing elastogram. A ROI was selected to produce the lateral shear wave speed profile. b) Lateral wave speed profile and sigmoid fit at the transition between soft and stiff region (n = 3 phantoms).

Figure 2.8 depicts an *ex vivo* porcine cornea under an IOP of 10 mmHg, where a strong reverberant field was induced within the cornea, as shown in the particle velocity maps at different time points (Figure 2.8b).

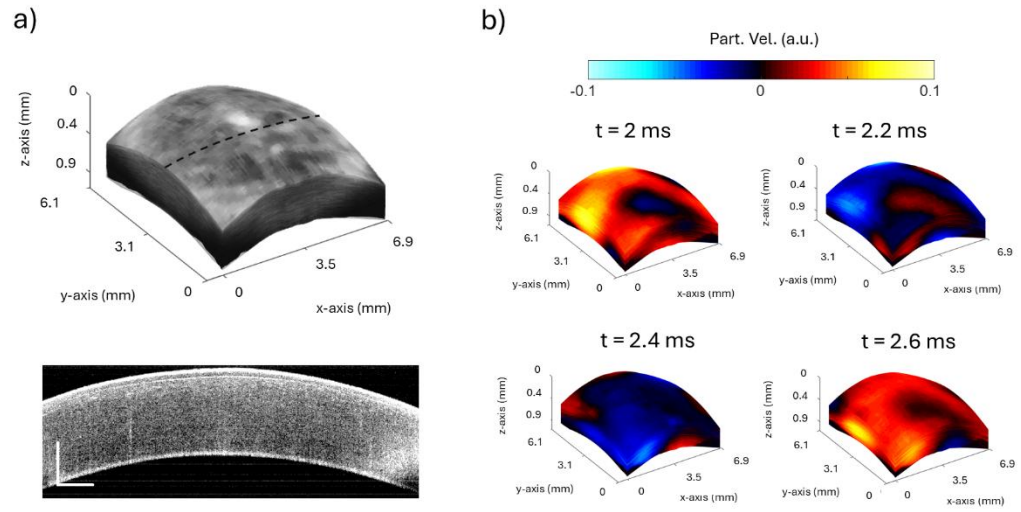


Figure 0.8. 3D OCT structural volume and B-mode image of the porcine cornea. b 3D volume snapshots of the reverberant shear field at the indicated time point after excitation.

The results show the depth dependence of the wave velocity from the epithelium to the endothelium in the cornea. Anterior corneal layers demonstrated higher velocity (3.7 m/s), i.e., higher stiffness, while the posterior layers were softer (2.2 m/s), shown on Figure 2.9.

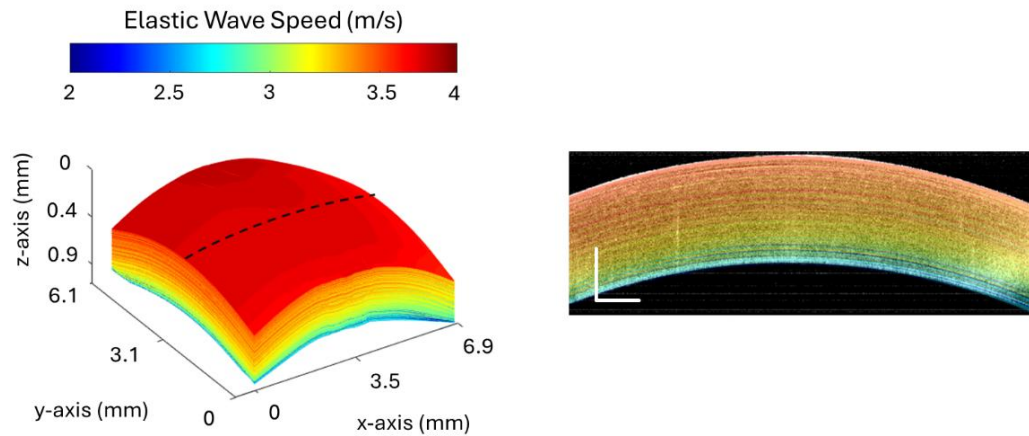


Figure 0.9. 3D volume of the shear wave speed map of the cornea and 2D wave speed slice.

2.4 Discussion

The acoustic radiation force generated by ACUS transducers arises from the significant impedance mismatch between air and tissue. Unlike immersion transducers, where acoustic energy penetrates deeply into the tissue and is primarily absorbed, most of the energy in air-coupled ultrasound is reflected at the tissue surface, generating radiation force. Consequently, air-coupled ultrasound is considered a safer method for tissue excitation compared to water-coupled ultrasound. Currently, no specific FDA safety regulations exist for air-coupled ultrasound; therefore, we based our estimations on the safety recommendations established for standard medical ultrasound in ophthalmology and an assumption of 99% energy reflectance at the tissue surface.

Since RevOCE was first shown, it has been applied to map the mechanical properties of various tissues, such as the cornea[155, 197], brain[198], and muscles [166], using contact-based excitation with an eight-pronged stimulator. To overcome the limitations of contact-based excitation, we have developed an ultrasound-based method to generate a reverberant field [172], but this method requires an acoustic coupling medium such as a gel or water. Thus, only the ACUS method is a truly “no touch” reverberant method. The present study demonstrates the promising generation of a robust reverberant field using commonly available ACUS transducers, and although they produce long diffuse ultrasonic beams over several meters, they can produce the acoustic radiation force at just 10 mm. These characteristics make them an ideal candidate to produce reverberant fields without any contact.

As shown in the Results section, a completely noncontact RevOCE methodology was developed, producing a strong reverberant wave propagation. As shown in the gelatin phantom (Figure 2.7) and an *ex vivo* porcine cornea (Figure 2.9), this method can be used to quantify the stiffness of tissues with high accuracy. Therefore, using a mechanically heterogeneous phantom was important to determine the RevOCE mechanical resolution that can be achieved using this methodology. There is a substantial difference in the amplitude of the wave propagation between 4% w/w and 8% w/w gelatin, which was previously observed in soft samples, where wave propagation seems to be diminished compared with the stiffer regions of the sample.

The 40 kHz transducers were chosen due to their broad commercial availability, cost-effectiveness (a pair of receiver-emitter transducers is ~\$1 USD), and lower resonant frequency as compared to previously demonstrated ACUS transducers (e.g., 1 MHz). This lower frequency enabled a robust and strong reverberant field. Higher frequencies are particularly useful in transient impulse wave-based OCE, where broadband elastic wave generation is critical for viscoelastic characterization [183]; however, this is not a strict requirement for RevOCE since multiple cycles of excitation are required in order to establish the reverberant field [167].

2.5 Conclusions

A completely noncontact method for RevOCE is a non-invasive technique used for OCE. This method can create a robust reverberant field within sample tissues, such as gelatin or sensitive tissues like corneas. Standard RevOCE techniques rely on M-B-

mode acquisition, which has the disadvantage of prolonged acquisition times. However, using B-M-mode acquisition protocols has shown a significantly lower time for data collection. This enhancement makes the method more suitable for potential clinical applications, while also minimizing discomfort for patients.

CHAPTER 3 – ADVANTAGES OF REVOCE COMPARED WITH A CONVENTIONAL UNIDIRECTIONAL WAVE-BASED OCE

Traditional wave-based OCE utilizes unidirectional excitation to measure the biomechanical properties of tissues. In this method, tissue characteristics such as anisotropy and heterogeneity play a crucial role in wave tracking. In contrast, RevOCE appears to benefit from tissue heterogeneity, as it induces increased reverberation of shear waves within the samples. This chapter compares RevOCE with traditional wave-based OCE in terms of mechanical resolution, using gelatin phantoms with varying boundary conditions.

3.1 Introduction

In wave-based OCE, boundary conditions significantly influence the interaction of mechanical waves with tissues, making them essential for accurate elasticity mapping [199]. However, excitation frequency, sample geometry, and complex boundary conditions can limit the elasticity resolution and contrast [200, 201]. In addition, the frequency of excitation significantly influences the mechanical resolution, where higher excitation leads to higher spatial resolution, allowing the detection of smaller and localized structures [202]. To overcome the limitations of traditional wave-based elastography, reverberant shear wave elastography was proposed, which leverages the interference of multiple waves traveling in random directions [154]. RevOCE overcomes the limitations of traditional wave-based elastography caused by natural tissue boundaries, providing superior elastography resolution and contrast [155, 166].

In addition, RevOCE promotes a higher proportion of shear wave propagation [203]. The studies about the frequency dependence of mechanical resolution in RevOCE are limited; however, some studies indicate that mechanical resolution in RevOCE also depends on frequency, like all other wave-based OCE methods. Therefore, increasing the excitation frequency reduces the shear wavelength, which can enhance imaging resolution and lead to more accurate elasticity maps [181]. In this chapter, the elastography resolution obtained using a traditional wave-based OCE method is compared with the resolution achieved using RevOCE in gelatin phantoms.

3.2 Materials and Methods

A Phs-SDOCT was utilized, using as light source a superluminescent diode (S480-B-I-20; Superlum Diodes Ltd., Ireland) with a central wavelength of 840 nm and a bandwidth of 50 nm, with an axial and transverse resolution of $\sim 9 \mu\text{m}$ and $\sim 8 \mu\text{m}$ in air, respectively [172].

A piezoelectric stack actuator (SA070742, Piezodrive, Australia) with a plastic tip attached to the end contacting the sample (Figure 3.1a) produced elastic waves in the samples using a quasi-harmonic signal of 5 pulses at 3 different frequencies (1, 5, and 10 kHz). The OCT system A-line speed was set to 25 kHz, 50 kHz, and 100 kHz for each excitation frequency (temporal resolution: Δt 25 kHz = 40 μs , Δt 50kHz = 20 μs , and Δt 100 kHz = 10 μs , respectively). Traditional wave-based OCE imaging was performed using M-B-mode imaging [183], where 1000 A-lines per position were imaged over 1000 positions across 4.74 mm for 1 kHz, 1.82 mm for 5 kHz, and 1.06 mm for 10 kHz. The wave propagation time and calculation of the local wave speed

were performed using previously described methods [149] using a kernel size of $1.5 \times 0.25 \text{ mm}^2$ at 1 kHz, $0.3 \times 0.1 \text{ mm}^2$ at 5 kHz, and $0.2 \times 0.08 \text{ mm}^2$ at 10 kHz, all in the xz-plane. In the case of RevOCE, the excitation source was a 3D-printed ring with eight rods attached to a piezoelectric bender (Figure 3.1b, PB4VB2W, Thorlabs Inc., NJ, USA) [155], and OCE imaging was performed using 3D-M-mode imaging by acquiring 101 by 101 M-mode points in the x and y axes, respectively, and 700 A-lines were acquired per position. The temporal resolution was the same as the traditional wave-based method at the corresponding frequency. In the case of 1 kHz, the scanning area was 11.41 mm along the x-axis and 13.46 mm for the y-axis. In the case of 5 and 10 kHz excitations, 5 blunt needles were attached in the internal diameter of the 3D printed ring, reducing the scanning area to 2.54 mm for the x-axis and 3.08 mm along the y-axis to generate a reverberant shear field with sufficient amplitude.

To evaluate the axial and lateral elastography resolutions, we utilized a heterogeneous phantom composed of 4% and 8% gelatin (w/w). This phantom, depicted in Figure 3.1c, featured distinct plane boundaries between its components, with different conformations (side-by-side, up-down, and L-shape). Three samples were assessed for each phantom configuration.

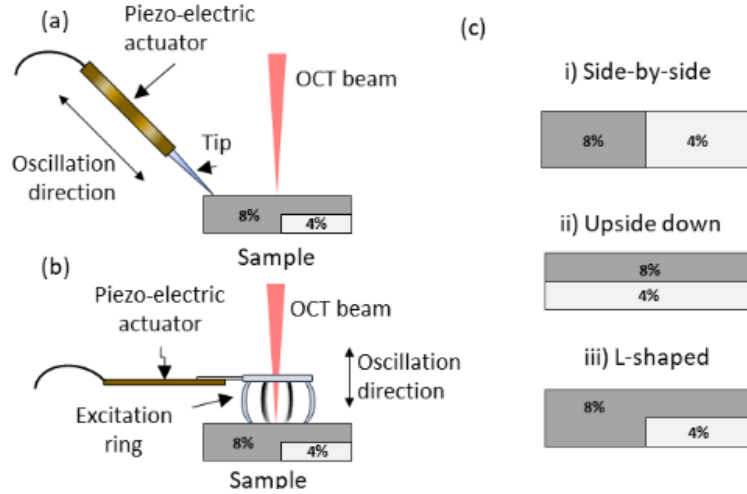


Figure 0.10. a) Generation of shear waves. b) Generation of the reverberant shear field. c) Heterogeneous gelatin phantoms (4% and 8%, w/w) in different arrangements: i) side-by-side, ii) up-down, and iii) L-shaped.

3.3 Results

Axial and lateral speed profiles were quantified and fitted to a Boltzmann sigmoid function. The full-width half-maximum derivative of the sigmoid was the mechanical resolution [172, 200]. Figure 3.2 illustrates the results of traditional wave-based elastography observed in a side-by-side gelatin phantom (left side was 8% gelatin phantom and the right side was 4% gelatin phantom). A quasi-harmonic excitation was applied to the sample using five pulses at frequencies of 1 kHz (Figure 3.2a-c), 5 kHz (Figure 3.2d-e), and 10 kHz (Figure 3.2g-i). The lateral elastography resolution was measured for each frequency.

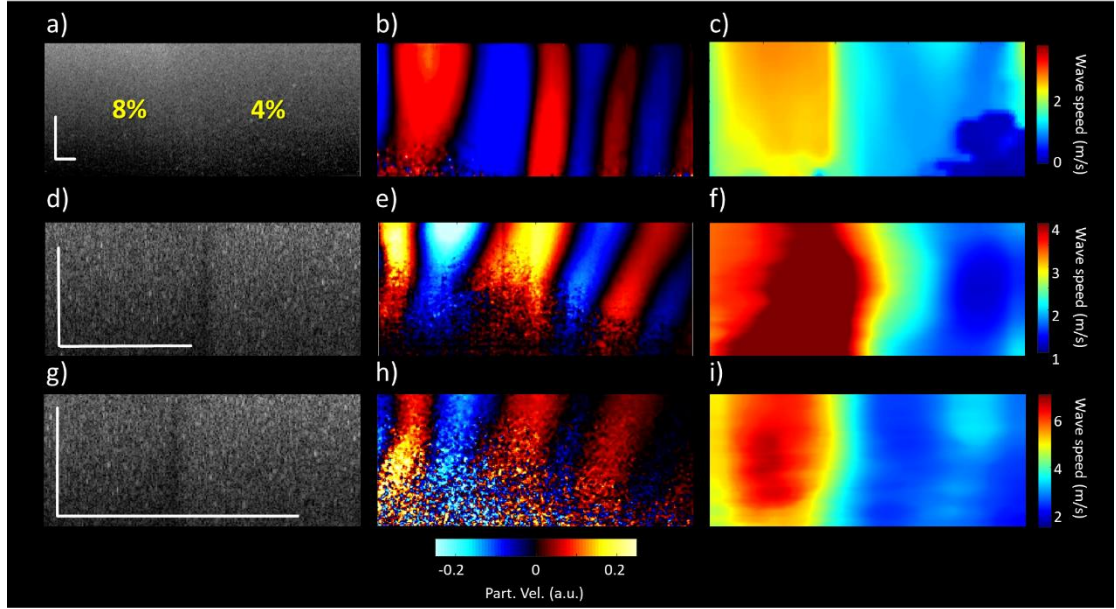


Figure 0.11. OCT images of the phantoms assessed at a) 1 kHz, d) 5 kHz, and g) 10 kHz. Wave propagation snapshots for b) 1 kHz, e) 5 kHz, and h) 10 kHz. Local wave speed maps at c) 1 kHz, f) 5 kHz, i) and 10 kHz. Scale bars are 1mm.

Furthermore, the axial elastography resolution in traditional wave-based elastography was evaluated using a gelatin phantom in an up-down (upper layer was 8% gelatin phantom and the lower layer was 4% gelatin phantom) arrangement (Figure 3.3). The applied excitation was a 5-pulse quasi-harmonic signal at frequencies of 1 kHz (Figure 3.3a-c), 5 kHz (Figure 3.3d-f), and 10 kHz (Figure 3.3g-i).

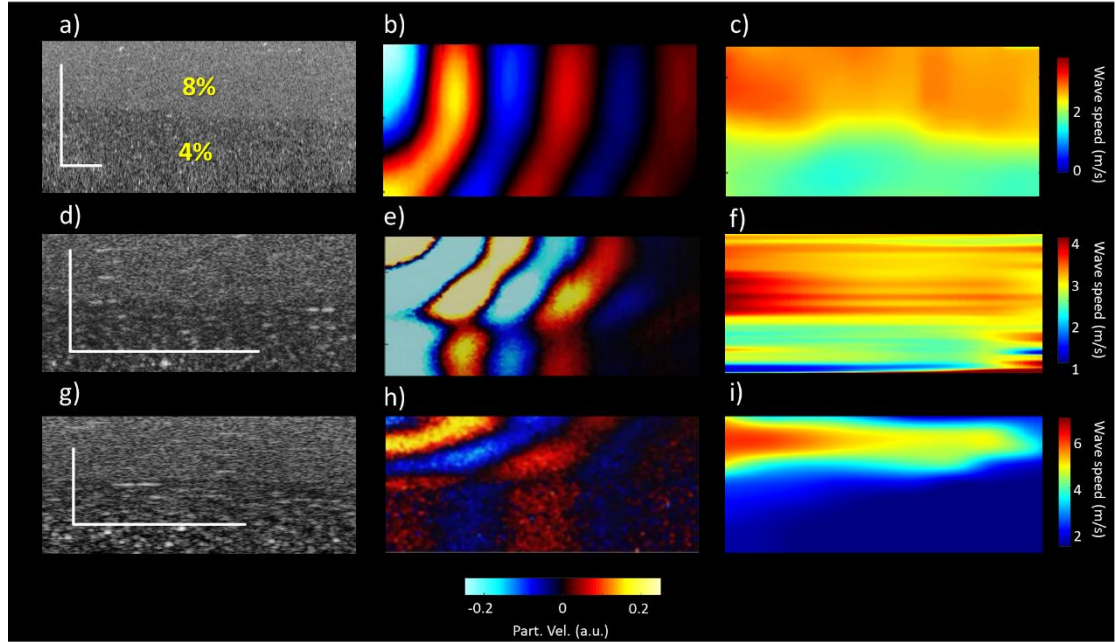


Figure 0.12. OCT images of the phantoms assessed at a) 1 kHz, d) 5 kHz, and g) 10 kHz. Wave propagation snapshots for b) 1 kHz, e) 5 kHz, and h) 10 kHz. Local wave speed maps at c) 1 kHz, f) 5 kHz, i) and 10 kHz. Scale bars are 1 mm.

For a more complex heterogeneous phantom, an L-shaped phantom consisting of an upper layer (8% gelatin phantom) and a lower layer (4% gelatin phantom), as shown in Figure 3.4. Similar to previous samples, an excitation of 5 pulses at frequencies of 1 kHz (Figure 3.4a-c), 5 kHz (Figure 3.4d-f), and 10 kHz (Figure 3.4g-i) were used in each case. The axial and lateral elastography resolution was quantified for all cases.

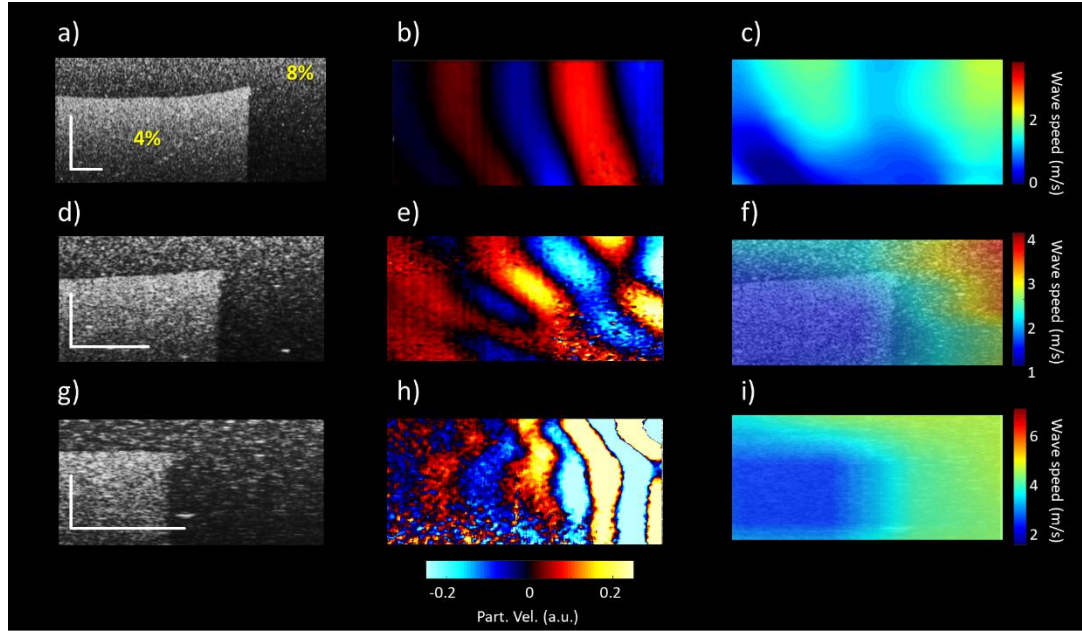


Figure 0.13. OCT images of the phantoms assessed at a) 1 kHz, d) 5 kHz, and g) 10 kHz. Wave propagation snapshots for b) 1 kHz, e) 5 kHz, and h) 10 kHz. Local wave speed maps at c) 1 kHz, f) 5 kHz, i) and 10 kHz. Scale bars are 500 μm

Selected motion frames of the quasi-harmonic excitation show the difference in the wave propagation at different stimulation frequencies (1, 5, and 10 kHz) for each type of phantom configuration. In the case of the side-by-side phantom, the particle velocity snapshots for the frequency of 1 kHz (Figure 3.2b), 5 kHz (Figure 3.2e), and 10 kHz (Figure 3.2h) show a clear difference in the wavelength between the gelatin phantom concentrations. In addition, in the case of up-down phantom configuration, the wave propagation shown in Figure 3.3b for 1 kHz, Figure 3.3e for 5 kHz, and Figure 3.3h for 10 kHz shows the differences in the wavelength between phantom layers. In contrast, the L-shaped phantom arrangement showed aberrant differences in the wave propagation (Figure 3.4) in this kind of phantom.

In terms of elastography resolution for traditional wave-based OCE (shown in Table 3.1), the lateral elastography resolution has been shown to be improved in side-by-side phantoms compared to L-shaped phantoms. In the case of the 1 kHz excitation, the resolution results showed that $199.2 \pm 38.3 \text{ }\mu\text{m}$ in the side-by-side phantom compared to $248.2 \pm 22.4 \text{ }\mu\text{m}$ obtained in the L-shaped phantom. On the other hand, when the excitation frequency was increased to 5 kHz, the lateral mechanical resolutions improved to $106 \pm 14.7 \text{ }\mu\text{m}$ and $152.4 \pm 33.4 \text{ }\mu\text{m}$ in the case of side-by-side and L-shaped phantoms, respectively. Furthermore, increasing the stimulation frequency to 10 kHz further improved the elasticity resolution to $77.9 \pm 16.6 \text{ }\mu\text{m}$ and $76.4 \pm 7.7 \text{ }\mu\text{m}$ in the case of side-by-side and L-shaped phantoms, respectively. In contrast, the axial elastography resolution obtained in the single-layer phantom (up-down phantom) showed a higher resolution compared to L-shaped phantoms, where, in the case of 1 kHz, the resolution was $162.9 \pm 5.2 \text{ }\mu\text{m}$ and $867.4 \pm 115.3 \text{ }\mu\text{m}$, respectively. The axial elastography resolution improved when the stimulation frequency of 5 kHz was applied, reducing to $64.3 \pm 8.4 \text{ }\mu\text{m}$ and $78.8 \pm 16.7 \text{ }\mu\text{m}$ in up-down and L-shaped phantoms, and when the frequency was increased to 10 kHz, a greater improvement was observed ($45.1 \pm 10.1 \text{ }\mu\text{m}$ and $50.8 \pm 21.4 \text{ }\mu\text{m}$ in up-down and L-shaped phantoms, respectively).

Table 0-1. Elastography resolution for traditional wave-based elastography across all phantom arrangements.

	Side-by-side phantom	Up-down phantom	L-shaped phantom	
Excitation Frequency (kHz)	Lateral (μm)	Depth (μm)	Lateral (μm)	Depth (μm)
1	199.2 ± 38.3	162.9 ± 5.2	248.2 ± 22.4	867.4 ± 115.3
5	106.3 ± 14.7	64.3 ± 8.4	152.4 ± 33.4	78.8 ± 16.7
10	77.9 ± 16.6	45.1 ± 10.1	76.4 ± 7.7	50.8 ± 21.4

In the case of RevOCE, to achieve a reverberant field with higher frequencies beyond 2 kHz (the highest frequency commonly used in RevOCE), it is important to consider the separation of the excitation sources while using high frequencies such as 5 kHz and 10 kHz. Due to wave amplitude damping, reducing the diameter of the excitation sources by frequency, as shown in Figure 3.5, was necessary.

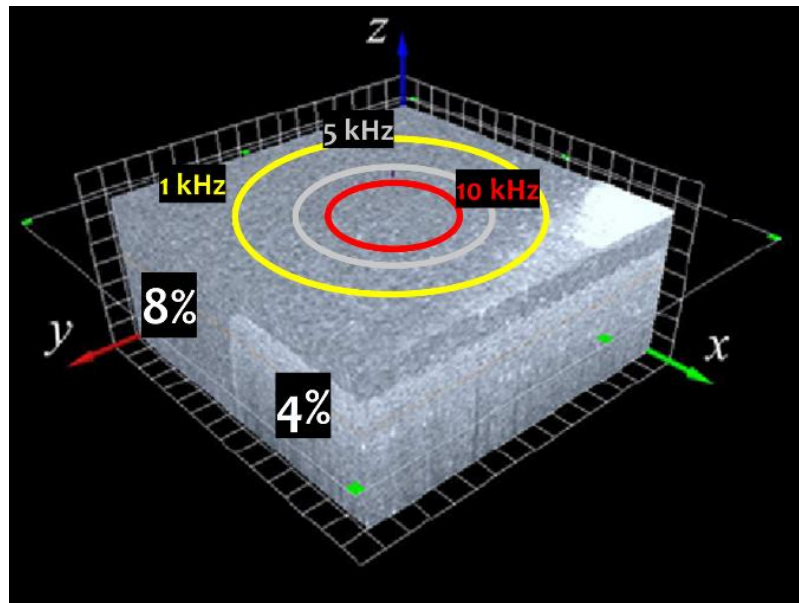


Figure 0.14. Distance of excitation sources for each frequency used in RevOCE.

To analyze the corresponding window sizes for frequencies of 1 kHz, 5 kHz, and 10 kHz for both 4% and 8% gelatin, different data processing window sizes were tested (see Figure 3.6).

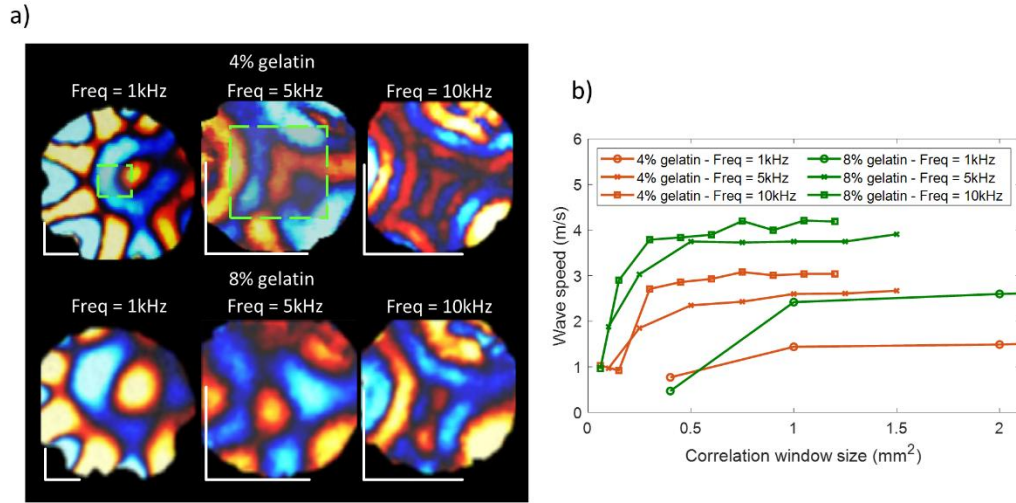


Figure 0.15. a) Particle velocity maps snapshots for each frequency (1 kHz, 5 kHz, and 10 kHz) for each gelatin concentration (4% and 8%). b) Wave speed of a given correlation window size for each gelatin concentration and frequency.

Figure 3.7 illustrates the results of RevOCE elastography observed in a side-by-side gelatin phantom (left side was 4% and the right side was 8%). A quasi-harmonic excitation was applied to the sample using ten pulses at frequencies of 1 kHz (Figure 3.7a-c), 5 kHz (Figure 3.7d-f), and 10 kHz (Figure 3.7g-i). The lateral elastography resolution was measured for each frequency.

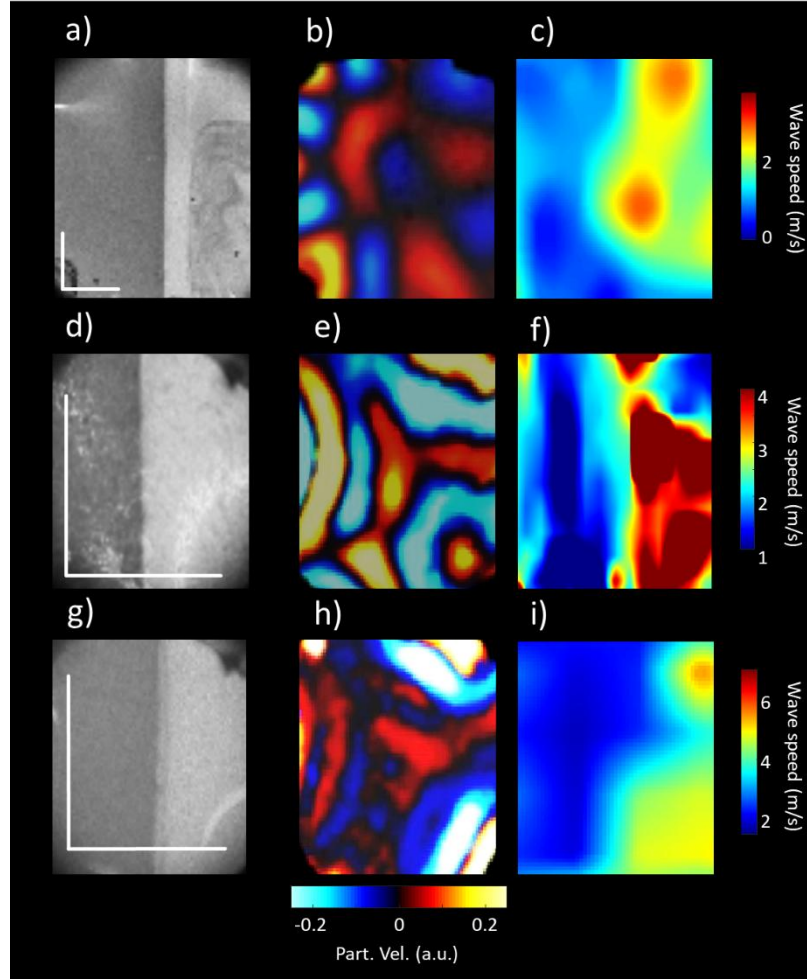


Figure 0.16. Images of side-by-side phantoms for a) 1 kHz, d) 5 kHz, and g) 10 kHz. RevOCE wave propagation snapshots for b) 1 kHz, e) 5 kHz, and h) 10 kHz. Local wave speed maps at c) 1 kHz, f) 5 kHz, i) and 10 kHz. Scale bar 1 mm.

Additionally, the axial elastography resolution in traditional wave-based elastography was evaluated using a gelatin phantom in an up-down (upper layer was 8% gelatin phantom and the lower layer was 4% gelatin phantom) arrangement (see Figure 3.8). The applied excitation was a 10-pulse quasi-harmonic signal at frequencies of 1 kHz (Figure 3.8a-c), 5 kHz (Figure 3.8d-f), and 10 kHz (Figure 3.8g-i).

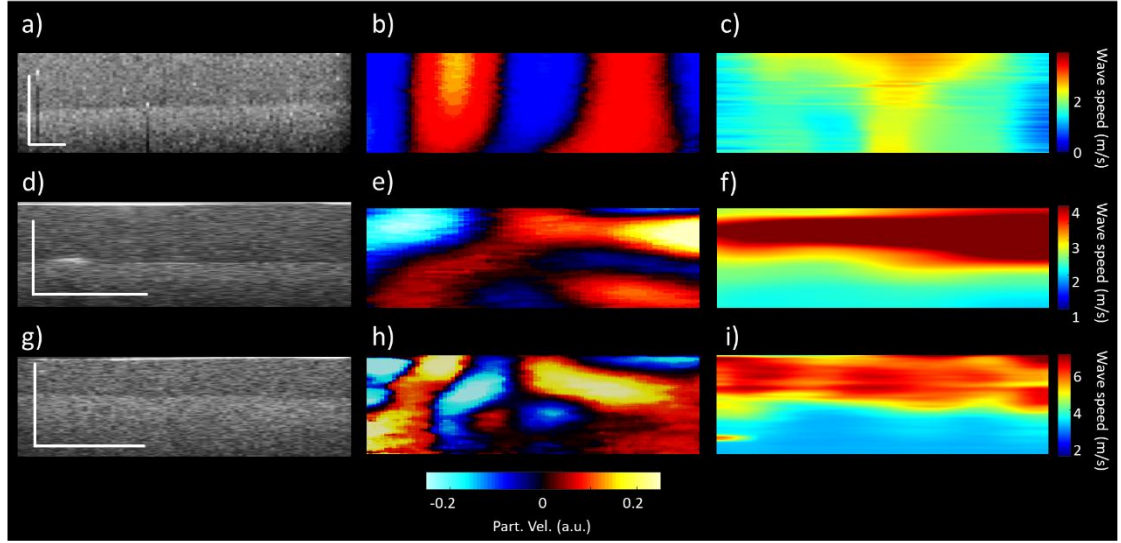


Figure 0.17. OCT images of up-down phantoms for a) 1 kHz, d) 5 kHz, and g) 10 kHz. RevOCE wave propagation for b) 1 kHz, e) 5 kHz, and h) 10 kHz. Local wave speed maps at c) 1 kHz, f) 5 kHz, i) and 10 kHz. Scale bar 1 mm.

In the complex heterogeneous phantom, an L-shaped phantom consisting of an upper layer (8% gelatin) and a lower layer (4% gelatin), as shown in Figure 3.9. Similar to previous samples, an excitation of 10 pulses at frequencies of 1 kHz (Figure 3.9a-c), 5 kHz (Figure 3.9d-f), and 10 kHz (Figure 3.9g-i) were used in each case. The axial and lateral elastography resolution was quantified for all cases.

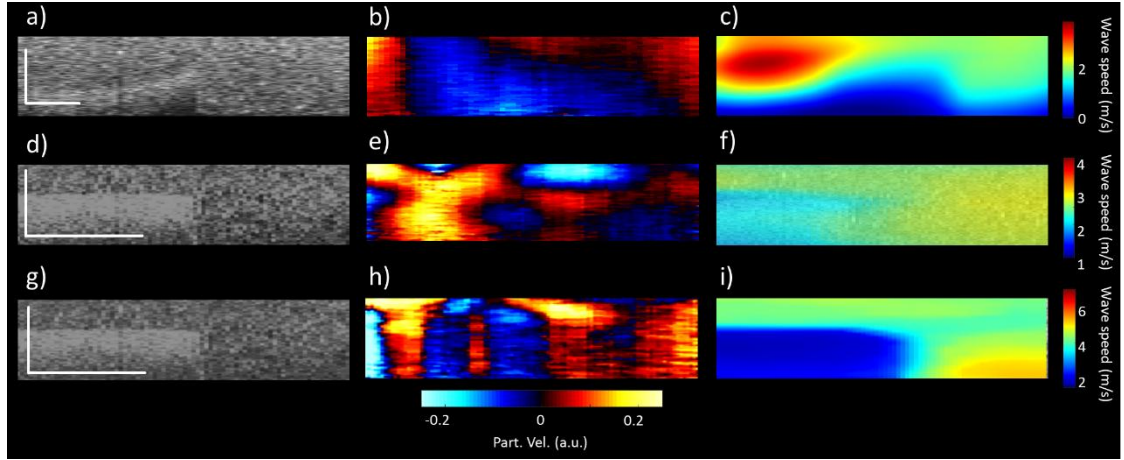


Figure 0.18. OCT images of L-shaped phantoms for a) 1 kHz, d) 5 kHz, and g) 10 kHz. RevOCE wave propagation for b) 1 kHz, e) 5 kHz, and h) 10 kHz. Local wave speed maps at c) 1 kHz, f) 5 kHz, i) and 10 kHz. Scale bar 1 mm.

The reverberant field was created by the shear waves generated by the excitation source. It is characteristic for each excitation frequency (1, 5, and 10 kHz) as shown in the side-by-side phantom (Figure 3.7), where the results showed a clear difference between the 4% gelatin phantom and the 8% gelatin phantom. In the case of up-down phantoms, the difference between layers was not clear at 1 kHz (Figure 3.8b), but improved when higher frequencies were used, such as 5 kHz (Figure 3.8e) and 10 kHz (Figure 3.8h). The same trend was observed in the results for the L-shaped phantoms, where the complex structure of this phantom at 1 kHz (Figure 3.9b) reduced the identification of the 4% layer; however, the results showed better discrimination of the layers at stimulation at frequencies 5 kHz (Figure 3.9e) and 10 kHz (Figure 3.9h).

The results for lateral and axial elastography resolution for RevOCE are summarized in Table 3.2. For lateral elastography resolution, the side-by-side phantom demonstrated a better resolution ($213.2 \pm 29.8 \mu\text{m}$) compared to the L-shaped phantom

($155.8 \pm 13.1 \mu\text{m}$) at a frequency of 1 kHz. The resolution improved when the frequency was increased to 5 kHz for the side-by-side phantoms ($72.5 \pm 12.3 \mu\text{m}$), while the resolution for the L-shaped phantoms was slightly lower at $81.1 \pm 9.5 \mu\text{m}$. At a frequency of 10 kHz, the resolutions were $46.2 \pm 17.1 \mu\text{m}$ for side-by-side phantoms and $65.4 \pm 10.5 \mu\text{m}$ for L-shaped phantoms. In terms of axial elastography resolution, the up-down phantom exhibited a lower resolution ($146.5 \pm 37.4 \mu\text{m}$) compared to the L-shaped phantom ($547.5 \pm 21.6 \mu\text{m}$). When the excitation frequency was increased to 5 kHz, the resolution improved to $53.5 \pm 10.5 \mu\text{m}$ for the up-down phantom and to $24.1 \pm 10.3 \mu\text{m}$ for the L-shaped phantom. Additionally, at 10 kHz, the resolutions further increased to $20.6 \pm 5.2 \mu\text{m}$ for the up-down phantom and to $16.6 \pm 6.5 \mu\text{m}$ for the L-shaped phantom.

Table 0.2. Elastography resolution for RevOCE elastography across all phantom configurations.

	Side-by-side phantom	Up-down phantom	L-shape phantom	
Excitation Frequency (kHz)	Lateral (um)	Depth (um)	Lateral (um)	Depth (um)
1	213.2 ± 29.8	146.5 ± 37.4	155.8 ± 13.1	547.5 ± 21.6
5	72.5 ± 12.3	53.5 ± 10.5	81.1 ± 9.5	24.1 ± 10.3
10	46.2 ± 17.1	20.6 ± 5.2	65.4 ± 10.5	16.6 ± 6.5

3.4 Discussion

The elastography resolution in OCE plays a crucial role in mapping and detecting tissue mechanical changes to detect diseases and monitor therapy efficacy [204]. In this study, traditional wave-based OCE and RevOCE showed a large difference in their elastography resolutions, specifically in the axial direction. As shown in the Results section, the lateral elastography resolution obtained by RevOCE is similar to wave-based OCE. However, in the case of axial elastography resolution, RevOCE was superior to traditional wave-based OCE. In both cases, lateral and axial resolution, a greater frequency increased the mechanical resolution, independent of the complexity of the gelatin phantom arrangement. Higher frequencies can provide sub-mm resolution, enabling the assessment of stiffer, smaller structures; however, lower frequencies allow deeper tissue penetration with lower resolution. Several factors affect the elastography resolution, such as the ability to detect the phase change to map wave propagation, and influence aspects such as signal-to-noise ratio (SNR), which is critical to capture accurate tissue displacements. [205].

The elastography resolution is directly related to the OCT imaging capabilities, where it is related to the capacity to track small induced mechanical deformation, and some specialized systems that achieve up to sub-2 μm resolution [206]. In wave-based OCE, the resolution of elastography is limited by the spatial wavelength of the mechanical wave. However, advanced techniques like RevOCE offer improved and more consistent elastography resolution. Therefore, this represents an advantage for

RevOCE to characterize the mechanical properties of tissues with complex geometries and varying properties [180].

3.5 Conclusions

In this chapter, the elastography resolution of RevOCE was superior to that of traditional wave-based OCE, particularly in complex phantom structures, e.g., the L-shaped phantoms. The results indicate that while traditional wave-based OCE exhibits high lateral and axial elastography resolution when boundary conditions are clearly defined, this resolution significantly diminishes in the presence of complex structures. This is where RevOCE shows marked improvements and shows that RevOCE may be particularly beneficial in cases of complex geometry, such as developing embryos.

CHAPTER 4 – USE OF REVOCE TO MAP THE MECHANICAL PROPERTIES OF DEVELOPING MOUSE EMBRYOS

During early embryonic development, significant biomechanical changes occur as the embryo grows. Understanding the structural modifications and variations in stiffness throughout this process is crucial for uncovering important growth mechanisms and identifying potential congenital diseases. The yolk sac serves as the primary source of nutrition for the fetus, where both micro- and macronutrients are absorbed through a process called pinocytosis. Disruption of yolk sac functions can lead to congenital malformations, impaired growth and development, and even embryonic death. In this chapter, reverberant optical coherence elastography (RevOCE) using multifocal acoustic radiation force (ARF) was used to map the stiffness of cultured mouse embryos within the yolk sac. Thus, this study provides important insights into the biomechanical properties of embryos, which could serve as a potential biomarker for detecting possible deformations during their development and revealing critical mechanisms involved in organogenesis. Since this elastography technique enabled high-resolution stiffness mapping in 3D with no need for external contrast agents or potentially damaging mechanical excitation, it has great potential for longitudinal imaging of embryos.

4.1 Introduction

Critical biomechanical changes occur in embryo growth during early embryonic development. The structural modifications and changes in stiffness throughout embryo growth are crucial for understanding its development and potentially detecting

congenital diseases. Majority of congenital diseases are structural and functional abnormalities that occur during embryo development, especially during the early days in the organogenesis, when major organs and structures are forming [207]. During embryogenesis, several fundamental biomechanical processes occur, such as elongation, cardiogenesis, and neural tube closure [208]. The yolk sac is the main support of embryo development and provides structural support before the placenta is fully functional, supporting all physiological processes[32]. In addition, the first hematopoietic cells circulated through the embryo are obtained from the yolk sac [209-211].

The yolk sac is the primary source of fetal nutrition, where micro and macronutrients are absorbed by pinocytosis [212] and receptor-mediated endocytosis [213]. The interference with the yolk sac functions could induce congenital malformations, deficient growth and development, and embryonic death. Furthermore, the abnormal growth of the yolk sac (larger or smaller) also represents the abnormal outcomes in embryo development and pregnancy outcomes, even representing physiological deficiencies. Thus, the normal yolk sac development represents a critical part of the progress of pregnancy [214]. In this work, reverberant optical coherence elastography (RevOCE) using multifocal ARF [172] was used to map the stiffness of mouse embryos within the yolk sac. The main aim of this study was to map the biomechanical properties of a mouse embryo at GD 8.5, 9.5, and 10.5 with high resolution in 3D.

4.2 Materials and Methods

The experiments were performed using mouse embryos at GD 8.5, 9.5, and 10.5 (n=5 per stage of development). The embryos were dissected while keeping the yolk sac intact. Embryo manipulations were performed at 37°C, and the embryos were conserved in an embryonic culture medium. The experimental setup was composed of a phase-sensitive optical coherence tomography (PhS-OCT) system with a superluminescent diode light source (Broadlighter S840-I-B-20, Superlum, Cobh Cross, Ireland) with a ~49 nm bandwidth and ~840 nm central wavelength. In the air, the lateral resolution was ~8 μm , and the axial resolution was ~9 μm . The A-line rate was set to 25 kHz (temporal resolution $\Delta t = 40 \mu\text{s}$). The PhS-SDOCT system was used for the detection of randomly distributed mechanical waves, which were produced by the multifocal ARF beams generated by an immersion ultrasound transducer (A392S-SU, Olympus Co., Japan) coupled to an acoustic lens array composed of plano-concave lenses (Figure 4.1). To induce reverberant shear wave fields in the embryos, an arbitrary function generator (DG4162, RIGOL Tech, China) produced a 1 MHz sinusoidal signal modulated by ten cycles of a 1 kHz rectangular tone burst, which was then amplified by a power amplifier (1040L, Electronics & Innovation, Ltd., United States) feeding the transducer. Three-dimensional RevOCE data were acquired with 500 A-lines per position, with a grid of 101 by 101 points acquired in total, resulting in a four-dimensional matrix of 3D space and time. In RevOCE, a reverberant field, which is the superposition of shear waves traveling in random directions, is generated within the sample. The reverberant field results in a motion perpendicular to the direction of the wave propagation.

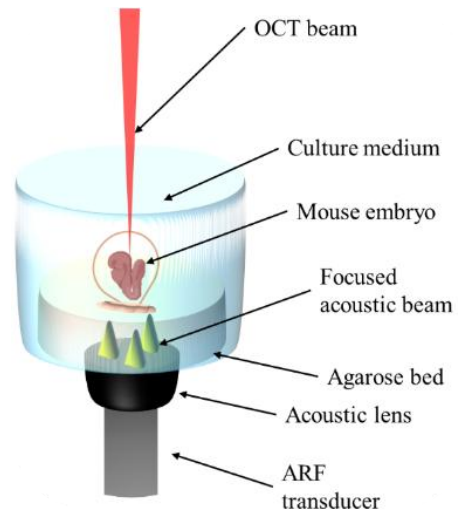


Figure 0.1. Schematic for RevOCE on mouse embryos. The embryos were placed inside the embryonic culture media while the OCE scanning was executed.

4.3 Results

The results showed the ability to map the structural differences among the mouse embryos, where the maximum intensity projections of 3D renderings of the embryos showed the maturation of embryos depicting three regions (heart, spine, and brain), as shown in Figure 4.2.

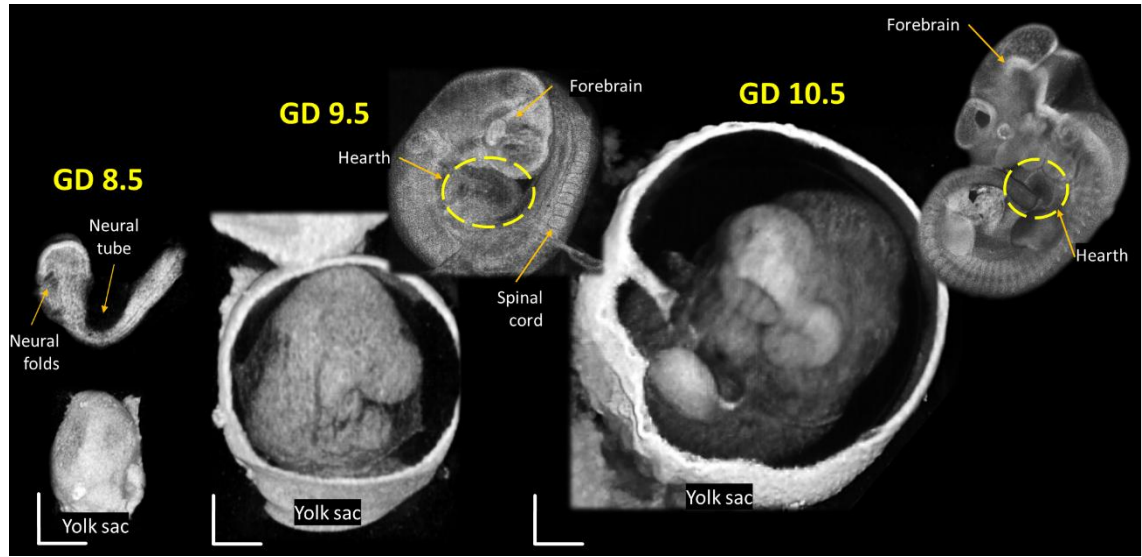


Figure 0.2. Maximum intensity projections and 3D OCT images of mouse embryos at GD 8.5, 9.5, and 10.5, illustrating the main parts of the embryo, including brain formation, heart, and spine. Scale bar 500 μm .

The particle velocity snapshots of the normalized out-of-plane reverberant shear field at the central sagittal plane of the embryos are shown in Figure 4.3. The heterogeneity of each embryo, which includes the yolk sac, has resulted in a randomly distributed reverberant shear field within the embryos.

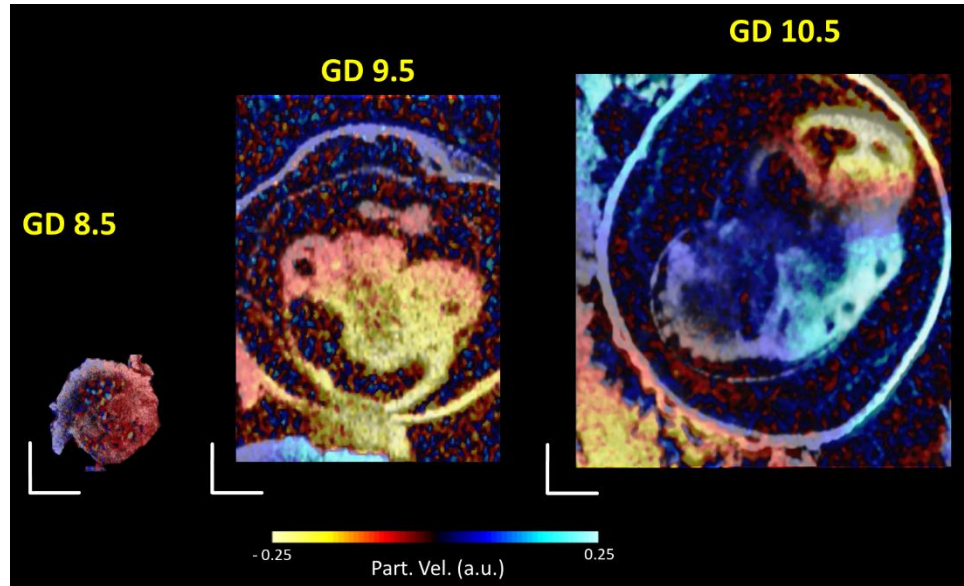


Figure 0.3. Snapshots depicting the particle velocity at a centrally located sagittal slice for each of the embryos (GD 8.5, 9.5, and 10.5), illustrating the velocity maps of mouse embryos within the yolk sac. Scale bar 500 μm .

The wave speed maps are shown in Figure 4.4, illustrating the stiffness of each embryo. The differences in stiffness among the mouse embryos are highlighted, particularly in this sagittal slice of the brain. Notably, the mouse embryo at GD 10.5 was stiffer compared to the other two embryos at GD 8.5 and 9.5, which is in agreement with previous work on mouse embryos removed from the yolk sac.

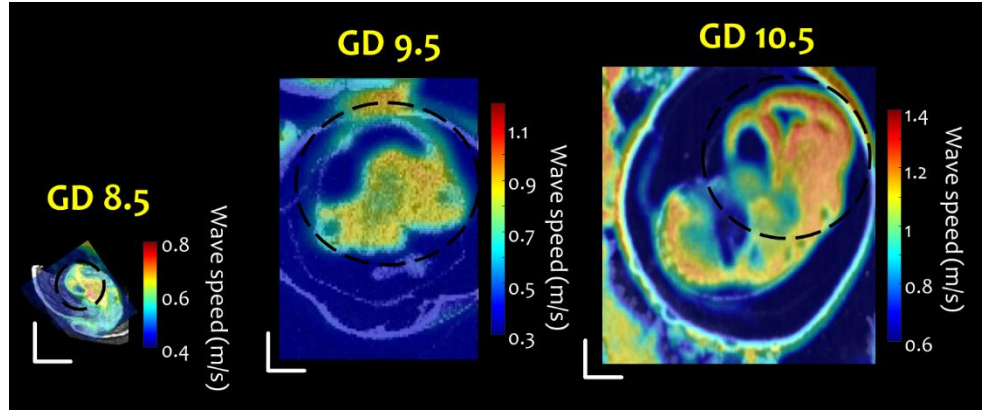


Figure 0.4. Local shear wave speed maps of each of the embryos. The highlighted region by a black dashed region represents the brain in each mouse embryo developmental stage.

The quantitative shear wave speed for three selected regions (brain, spine, and heart) was quantified for each embryo at different developmental stages, as shown in Figure 4.5. A Kruskal Wallis ANOVA showed a significant effect of the development stage on the wave speed in the brain ($\chi^2 = 12.02$, $DF = 14$, $P = 0.0025$), spine ($\chi^2 = 9.5$, $DF = 14$, $P = 0.0087$), and heart ($\chi^2 = 12.32$, $DF = 14$, $P = 0.0021$).

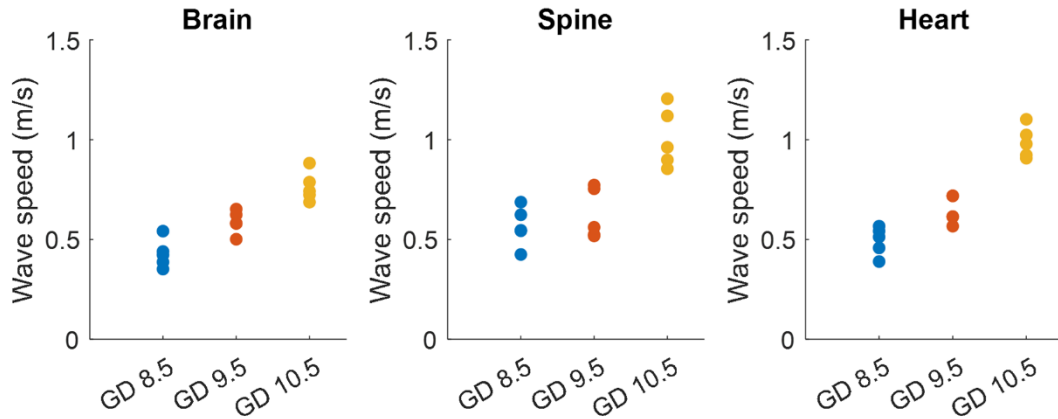


Figure 0.5. Local wave speed average for each embryo of various developmental stages. The three regions (spine, heart, and brain) analyzed for each mouse embryo was analyzed in each case, and $n=5$ embryos per stage of development.

4.4 Discussion

The biomechanical properties of the embryos govern the development of tissues and organs, and they are a key factor during their development [215]. The failure of these mechanical cues in the normal embryo development produces significant deficiencies in the embryo structure and even the completion of the gestational process. Thus, monitoring and measuring the biomechanical properties of embryos as they develop is essential for understanding these processes. Previous methods have been used to assess the mechanical properties of mouse embryos, such as AFM [216] and optical tweezers [217]. Another optical elastography method, Brillouin microscopy, has been used to measure the biomechanical properties of embryos to assess the neural tube closure; however, this method cannot obtain the embryo structure while it is developing [40] and lacks speed for 3D imaging within reasonable time frames (one 2D image requires ~30 minutes). In addition, previously, RevOCE was used to map the biomechanical properties of mouse embryos from GD 9.5 to 11.5, where the advantage of using OCT's high resolution imaging capabilities was combined with the elastography advantages of RevOCE, which overcomes the limitation of traditional wave-based OCE in samples with highly complex boundary conditions and enables noninvasive mapping of the stiffness of whole embryos while they develop. However, this method demands that mouse embryos be completely dissected, without a yolk sac, preventing longitudinal analysis of the biomechanical properties of embryos while they develop.

Using the multifocal ARF method to analyze the biomechanical properties of embryos while they remain intact in the yolk sac potentially overcomes the limitations observed in the previous elastography methods [172]. This method allowed the evaluation of elastic wave speed in different parts of the embryos at different developmental stages. For instance, embryos at a GD 10.5 show a regional variation in stiffness, where the spine becomes stiffer compared to the rest of the embryos at different stages. In addition, there is a significant stiffness difference in the head development, where at GD 8.5 [218], the neural tube folds to form the brain and central nervous system. Later, at a stage of GD 9.5, main events such as the neural tube closure are completed, and the cranial nerves and secondary brain vesicles are formed at a stage of GD 10.5 [219].

4.5 Conclusions

Combining high-resolution OCT with RevOCE using multifocal ARF enabled mapping of the structural and biomechanical properties of murine embryos with high resolution in 3D while the embryo was inside the intact yolk sac. This work shows the capability of RevOCE to characterize biomechanical properties of the embryo during its development noninvasively with no need for exogenous contrast agents either for structural imaging or biomechanical mapping, which shows promise for longitudinal imaging of embryo development.

CHAPTER 5 – USE OF REVOCE TO ASSESS THE BIOMECHANICAL PROPERTIES OF EMBRYO AND LARVAL ZEBRAFISH

During embryonic development, the tissue biomechanical properties shape the formation of organs and systems, from the initial cellular stages to the formation of the organs and whole bodies. Zebrafish is an animal model whose optically clear *ex-utero* embryo development makes it suitable for biomedical imaging research from the initial hours post-fertilization until the complete formation of the zebrafish. In this chapter, the biomechanical properties of zebrafish embryos and larval development are studied from the initial stages (~ 1 hpf), where initial cellular differentiation starts, until a complete zebrafish around 5 days post fertilization.

5.1 Introduction

Zebrafish are a preferred model in many biomedical applications due to their high transparency, allowing real-time and non-invasive imaging, which makes fluorescence imaging easy [220]. However, zebrafish are not entirely transparent and have pigments that develop within the first 2 to 5 dpf. The pigmentation begins in the neural crest during the embryonic stage, where the dermal melanophores respond to light, triggering pigment dispersion, which later will develop into the skin stripes commonly observed in zebrafish from 3 weeks to 90 days [221]. Transparency is an advantage that has been used in OCT, where it was used to study the development of zebrafish from early embryonic stages (0 hpf to 3 dpf) to larval and juvenile stages and

adult stages. During these early stages, OCT has been used to study ocular, cardiovascular, and neurological development [222, 223].

Several methods have been used to study the biomechanical properties of zebrafish embryos, such as Brillouin microscopy [63]. The biomechanical changes that zebrafish embryos undergo during their development have been studied previously using mechanical models to explore the changes during the formation of the somites [224]. Thus, measuring the biomechanical properties of zebrafish embryos and larvae is important to understand their development. The main aim of this study was to map the biomechanical properties of zebrafish embryos at 1 hpf, 24 hpf, 48 hpf, and 5 dpf with high resolution in 3D.

5.2 Materials and Methods

The experiments were conducted using *in vivo* wild-type (WT) zebrafish at 1 hpf, 24 hpf, 48 hpf, and 5 dpf. The embryos were cultured at 29°C and were kept alive in fish water while they developed. The embryos were imaged using 1% agarose plate, where the reverberant field was induced to propagate waves towards the zebrafish embryos [175]. The zebrafish embryos were placed on the agarose surface, and the reverberant waves were induced using a 3D printed ring, which was connected to a piezoelectric actuator (Thorlabs, PB4VB2W) driven by a high-voltage amplifier (Piezodrive, MX200) as shown on Figure 5.1. RevOCE measurements were made with a phase-sensitive optical coherence tomography (PhS-OCT) system described in the previous chapters. During the RevOCE measurements, the A-line acquisition speed was set to 25 kHz (temporal resolution $\Delta t = 40 \mu s$). The window also enabled common path

imaging for sub-nanometer displacement sensitivity. A 1 kHz sinusoidal signal produced by a function generator (DG4162, RIGOL Tech, Beijing, China) was sent to an amplifier before it drove the bender. Three-dimensional RevOCE data was acquired with 500 A-lines per position, with a grid of 151 by 151 points acquired in total, resulting in a four-dimensional matrix of 3D space and time. The X-axis scan was 4.52 mm, and the Y-axis scan was 1.62 mm.

Using the spatial and temporal data from the acquired 4D OCE data, a 3D wave speed map was reconstructed using RevOCE processing methods [155]. In RevOCE, the interference of shear waves traveling in random directions generates a reverberant field. Two-dimensional spatial autocorrelation profiles (window size: 0.4 mm by 0.4 mm in the X and Y directions for a given Z plane) were calculated and fitted to the analytical solutions from the reverberant shear wave speed model to obtain the local wave number (k) and subsequently, the local shear wave speed (c_s) by Equation 1.2. This process was repeated for each depth imaged in the samples.

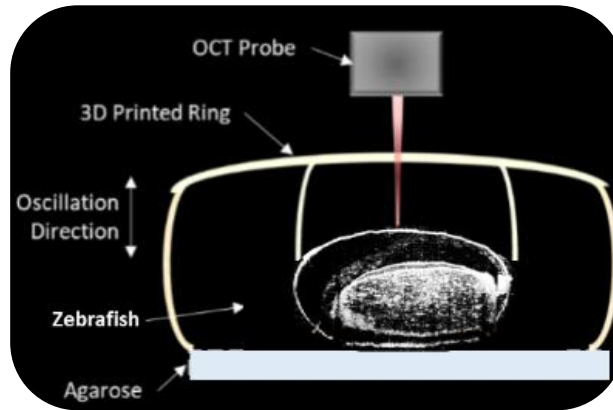


Figure 0.1. Schematic for RevOCE on zebrafish embryos. The embryos were placed on top of the agarose dish media while the OCE scanning was executed.

5.3 Results

The results showed the ability of OCT to image the zebrafish development from early stages (1 hpf) to later days of development (5 dpf). As shown in Figure 5.2, the zebrafish embryos develop from early stages, where the zebrafish embryos are inside the chorion, and later stages, where, at 24 hpf to 48 hpf, the zebrafish embryos break the chorion and start their maturation in later days, such as 5 dpf.

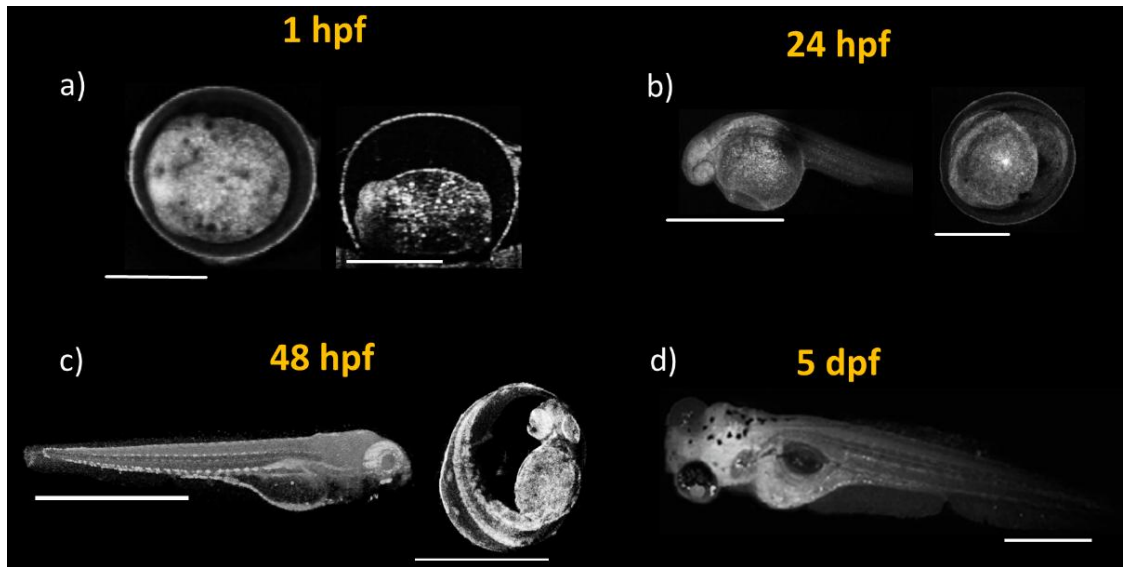


Figure 0.2. Maximum intensity projections of 3D OCT images of zebrafish embryos at 1 hpf, 24 hpf, 48 hpf, and 5 dpf. Scale bar 1 mm.

The particle velocity snapshots of the normalized out-of-plane reverberant shear field at the central sagittal plane of the zebrafish embryos are shown in Figure 5.3. The heterogeneity of each embryo has resulted in a randomly distributed reverberant shear field.

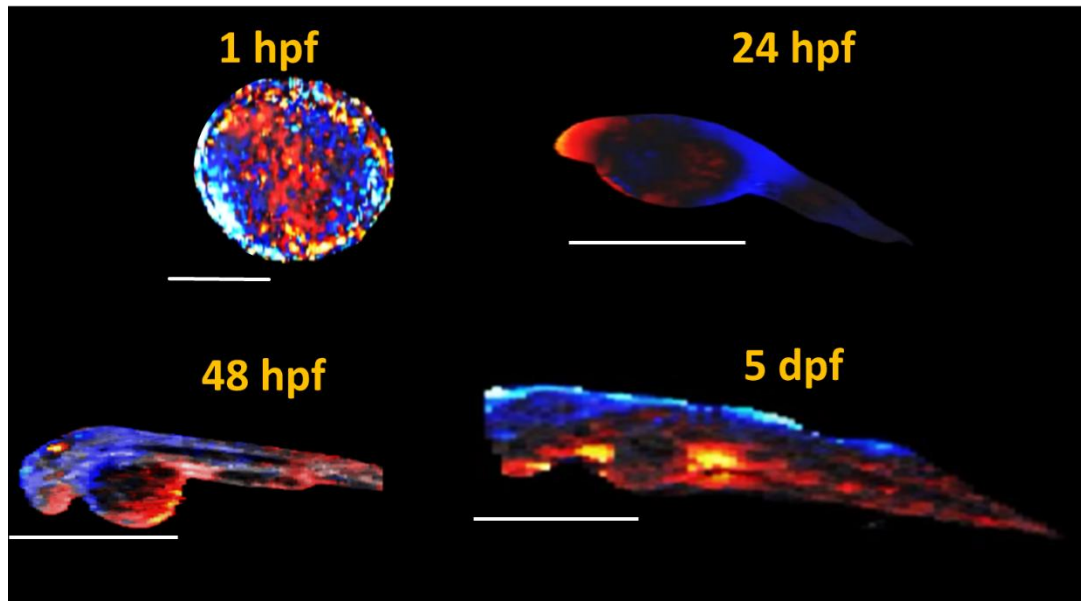


Figure 0.3. Snapshots depicting the particle velocity at a centrally located sagittal slice for each of the embryos (1 hpf, 24 hpf, 48 hpf, and 5 dpf), illustrating the velocity maps of mouse embryos within the chorion. Scale bar 1 mm.

The wave speed maps illustrate the stiffness differences among the zebrafish embryos, as shown in Figure 5.4, where a notable difference in stiffness is observed from 1 hpf to 5 dpf. Particularly, the zebrafish embryos at 24 hpf and 48 hpf showed that stiffness is related to their heads and spines.

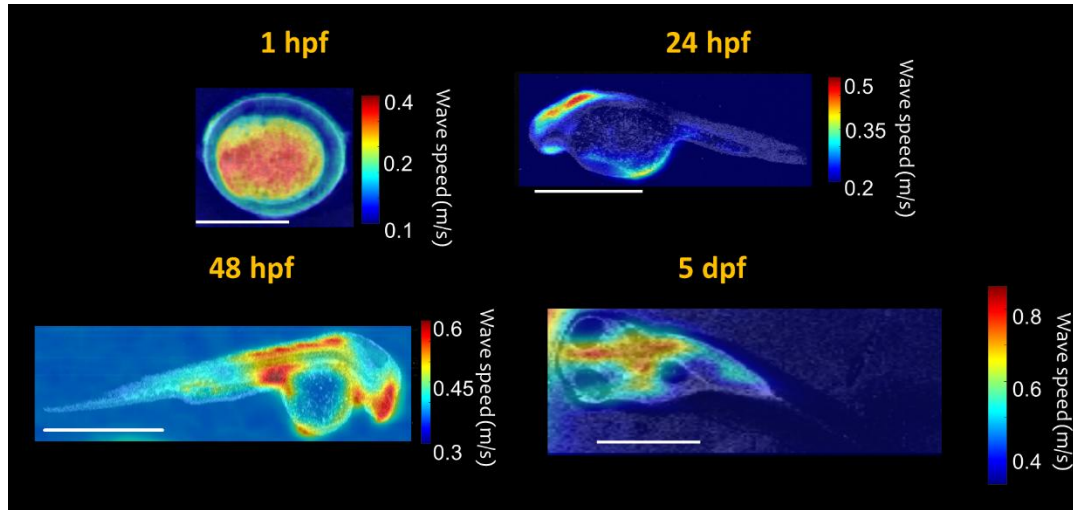


Figure 0.4. Local shear wave speed maps of each of the zebrafish embryos in each developmental stage (1 hpf, 24 hpf, 48 hpf, and 5 dpf).

The quantitative shear wave of the whole body was quantified for each zebrafish embryo at different developmental stages shown on Figure 5.5. Using Kruskal Wallis ANOVA showed a significant effect of the development on the shear wave speed ($\chi^2 = 9.97$, $DF = 11$, $P = 0.0188$).

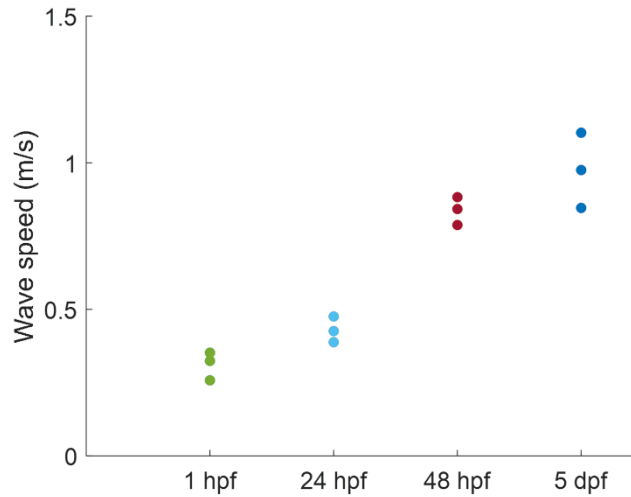


Figure 0.5. Local shear wave speed average for each zebrafish embryo of various developmental stages. The whole-body speed was analyzed for each zebrafish embryo was analyzed in each case (n=3).

5.4 Discussion

Similar to what was previously discussed in Chapter 4, the biomechanical properties of tissues play an important role in the development of zebrafish embryos. One of the main advantages of zebrafish over mouse embryos is the ease with which one can monitor very early stages of embryonic development. Thus, zebrafish is an ideal animal model for the early detection of diseases and genetic manipulation [225]. Indeed, several diseases, such as craniofacial [226] and neural tube defects [227], are extensively studied using zebrafish. In later stages, which are reached at 24 hpf, important morphological features are completed, such as the head, the eyes, and the tail, which are well-structured. In addition, the spontaneous muscle motility begins [228]. During the next 24 hpf, the zebrafish hatch from the chorion, and the internal organs are completely developed, and at 5 dpf the fish transition completely from embryo to a free-

feeding larva [229]. During all this time, as shown in the result section, the biomechanical properties of tissues show that the stiffness increases progressively.

As shown in Figure 5.2, during their development, zebrafish had several anatomical changes, where they transform from an oval shape at the early embryonic stage to an enlarged fish shape at the larval stage. Because of this, zebrafish need to be mounted for their image in similar orientation conditions. 3D-printed stamps that create micro wells on agarose facilitate the imaging of zebrafish embryos simultaneously; however, this method is not uniform for all stages, from early stages up to the larval stage [230, 231]. Thus, the agarose bed was used in this study to easily manipulate the zebrafish in different stages during their embryonic development.

5.5 Conclusions

This chapter discusses the results of RevOCE to map the biomechanical properties of zebrafish embryos. OCT structural images showed that OCT can monitor the development of zebrafish from early stages, such as 1 hpf, to late stages, like 5 dpf. Similar to chapter 5, this study also demonstrates a link between the biomechanical properties measured by RevOCE and the stiffness of zebrafish embryos, where RevOCE was able to map the stiffness of zebrafish embryos in 3D with high resolution and showed that embryos increased their stiffness during their development.

CHAPTER 6 – CONCLUSIONS

In this dissertation, RevOCE is analyzed and implemented in a completely noncontact version, and its higher capacity for elastography resolution is utilized to study the biomechanical characteristics of zebrafish and mouse embryo development.

There are four major sections in this dissertation, which are described below:

1. RevOCE was developed in a contact manner using a 3D printed ring that generated a strong reverberant field within the samples. Later, an immersion ARF transducer was used to induce a reverberant field as a noncontact method. However, this approach needed a coupling medium to induce acoustic waves in the sample. A completely noncontact method with the capacity to produce a high-amplitude reverberant field was needed. Inexpensive ACUS transducers were utilized to perform RevOCE, and the results showed generation of a robust reverberant shear field within a variety of samples, including an *ex vivo* porcine cornea. This study presents proof of concept for the feasibility of a completely noncontact method to study the biomechanical properties of tissues with high resolution and contrast.

2. Traditional wave-based OCE uses a unidirectional excitation to determine the biomechanical properties of tissues. However, complex boundary conditions are a disadvantage for this method, because tissue heterogeneity can generate distortions in the wave propagation and reduce the accuracy of speed calculations. In contrast, RevOCE benefits from heterogeneous tissues because

it can increase the wave reverberations within the sample. In this study, the results showed that RevOCE possesses higher axial elastography resolution compared to traditional wave-based OCE. Furthermore, higher frequencies improve the RevOCE capacity to resolve the biomechanical properties of tissues in smaller areas.

3. The biomechanical properties of embryos are an important marker for their normal development. Hence, differences in the mechanical properties of embryos can indicate the appearance of abnormalities such as craniofacial and neural tube defects. Several imaging modalities have been utilized to obtain the biomechanical properties of embryos; however, they cannot map the entire embryo's mechanical properties in 3D noninvasively and in a reasonable time, which is needed for longitudinal analysis of embryo development. Here, an immersion ARF-based RevOCE technique was utilized to measure the biomechanical properties of whole embryos inside the yolk sac at different developmental stages.

4. Zebrafish embryos offer the capacity to observe the development of embryos from the initial stages immediately after fertilization. The initial embryo stages, which include a reduced number of cells, and the complete development of tissues and organs in later zebrafish stages, are visible during their development. In this study, RevOCE was used to monitor the biomechanical properties of zebrafish embryos during their development. This method offers an opportunity to observe not only the zebrafish development but

also the biomechanical properties from early stages of embryo development (1 hpf) up to several days after fertilization (5 dpf).

The impact of this dissertation is to develop and validate a completely noncontact RevOCE approach, confirm RevOCE's superior elastography resolution in comparison to traditional wave-based OCE, and showcase RevOCE's use in developmental biology. The work presented in this dissertation demonstrates the link between the developmental stages of mouse and zebrafish embryos and the increase in their local stiffness, and it can progress in the study of new therapeutic interventions to detect and prevent defects while embryos are developing.

REFERENCES

- [1] K. Goodwin and C. M. Nelson, "Mechanics of Development," (in eng), *Dev Cell*, vol. 56, no. 2, pp. 240-250, Jan 25 2021, doi: 10.1016/j.devcel.2020.11.025.
- [2] P. Agarwal and R. Zaidel-Bar, "Mechanosensing in embryogenesis," (in eng), *Curr Opin Cell Biol*, vol. 68, pp. 1-9, Feb 2021, doi: 10.1016/j.ceb.2020.08.007.
- [3] E. Paluch and C. P. Heisenberg, "Biology and physics of cell shape changes in development," (in eng), *Curr Biol*, vol. 19, no. 17, pp. R790-9, Sep 15 2009, doi: 10.1016/j.cub.2009.07.029.
- [4] I. Pajic-Lijakovic and M. Milivojevic, "Physics of collective cell migration," (in eng), *Eur Biophys J*, vol. 52, no. 8, pp. 625-640, Nov 2023, doi: 10.1007/s00249-023-01681-w.
- [5] W. Piszker and M. Simunovic, "The fusion of physics and biology in early mammalian embryogenesis," (in eng), *Curr Top Dev Biol*, vol. 160, pp. 31-64, 2024, doi: 10.1016/bs.ctdb.2024.05.001.
- [6] C. Villeneuve, K. P. McCreery, and S. A. Wickström, "Measuring and manipulating mechanical forces during development," (in eng), *Nat Cell Biol*, vol. 27, no. 4, pp. 575-590, Apr 2025, doi: 10.1038/s41556-025-01632-x.
- [7] K. H. Vining and D. J. Mooney, "Mechanical forces direct stem cell behaviour in development and regeneration," (in eng), *Nat Rev Mol Cell Biol*, vol. 18, no. 12, pp. 728-742, Dec 2017, doi: 10.1038/nrm.2017.108.

- [8] J. Petzold and E. Gentleman, "Intrinsic Mechanical Cues and Their Impact on Stem Cells and Embryogenesis," (in eng), *Front Cell Dev Biol*, vol. 9, p. 761871, 2021, doi: 10.3389/fcell.2021.761871.
- [9] N. Tarannum, R. Singh, and S. Woolner, "Sculpting an Embryo: The Interplay between Mechanical Force and Cell Division," (in eng), *J Dev Biol*, vol. 10, no. 3, Sep 1 2022, doi: 10.3390/jdb10030037.
- [10] R. E. Poelmann, A. C. Gittenberger-de Groot, and B. P. Hierck, "The development of the heart and microcirculation: role of shear stress," (in eng), *Med Biol Eng Comput*, vol. 46, no. 5, pp. 479-84, May 2008, doi: 10.1007/s11517-008-0304-4.
- [11] H. Zhu, J. Guo, H. Wang, D. Gu, D. Wang, and Y. Liu, "Developmental changes of free amino acids in amniotic, allantoic fluids and yolk of broiler embryo," (in eng), *Br Poult Sci*, vol. 63, no. 6, pp. 857-863, Dec 2022, doi: 10.1080/00071668.2022.2079073.
- [12] L. Hu, W. Chen, A. Qian, and Y. P. Li, "Wnt/ β -catenin signaling components and mechanisms in bone formation, homeostasis, and disease," (in eng), *Bone Res*, vol. 12, no. 1, p. 39, Jul 10 2024, doi: 10.1038/s41413-024-00342-8.
- [13] R. A. Taft, "Virtues and limitations of the preimplantation mouse embryo as a model system," (in eng), *Theriogenology*, vol. 69, no. 1, pp. 10-6, Jan 1 2008, doi: 10.1016/j.theriogenology.2007.09.032.

- [14] E. C. Bryda, "The Mighty Mouse: the impact of rodents on advances in biomedical research," (in eng), *Missouri medicine*, vol. 110, no. 3, pp. 207-11, May-Jun 2013.
- [15] M. B. Veldman and S. Lin, "Zebrafish as a developmental model organism for pediatric research," (in eng), *Pediatric research*, vol. 64, no. 5, pp. 470-6, Nov 2008, doi: 10.1203/PDR.0b013e318186e609.
- [16] C. Santoriello and L. I. Zon, "Hooked! Modeling human disease in zebrafish," (in eng), *The Journal of clinical investigation*, vol. 122, no. 7, pp. 2337-43, Jul 2012, doi: 10.1172/jci60434.
- [17] N. Virina, L. Kuchieva, Y. Baturina, A. Fizikova, M. Gereeva, B. Bitiev, K. Apakhaeva, N. Manukhova, F. Rasulova, E. Kornev, and E. Rodionova, "DANIO RERIO (ZEBRAFISH) - A UNIQUE AND INTEGRATIVE PLATFORM FOR 21ST CENTURY BIOMEDICAL RESEARCH," (in eng), *Georgian medical news*, no. 369, pp. 168-173, Dec 2025.
- [18] M. Volkov, A. Machikhin, V. Bukova, D. Khokhlov, A. Burlakov, and V. Krylov, "Optical transparency and label-free vessel imaging of zebrafish larvae in shortwave infrared range as a tool for prolonged studying of cardiovascular system development," (in eng), *Scientific reports*, vol. 12, no. 1, p. 20884, Dec 3 2022, doi: 10.1038/s41598-022-25386-w.
- [19] S. Ye, R. Yang, J. Xiong, K. K. Shung, Q. Zhou, C. Li, and Q. Ren, "Label-free imaging of zebrafish larvae in vivo by photoacoustic microscopy," (in eng),

- Biomedical optics express*, vol. 3, no. 2, pp. 360-5, Feb 1 2012, doi: 10.1364/boe.3.000360.
- [20] Y. Tian, Y. Zhan, Q. Jiang, W. Lu, and X. Li, "Expression and function of PDGF-C in development and stem cells," (in eng), *Open biology*, vol. 11, no. 12, p. 210268, Dec 2021, doi: 10.1098/rsob.210268.
- [21] C. Qiu, B. K. Martin, I. C. Welsh, R. M. Daza, T. M. Le, X. Huang, E. K. Nichols, M. L. Taylor, O. Fulton, D. R. O'Day, A. R. Gomes, S. Ilcisin, S. Srivatsan, X. Deng, C. M. Disteché, W. S. Noble, N. Hamazaki, C. B. Moens, D. Kimelman, J. Cao, A. F. Schier, M. Spielmann, S. A. Murray, C. Trapnell, and J. Shendure, "A single-cell time-lapse of mouse prenatal development from gastrula to birth," (in eng), *Nature*, vol. 626, no. 8001, pp. 1084-1093, Feb 2024, doi: 10.1038/s41586-024-07069-w.
- [22] H. Li, L. Chang, J. Wu, J. Huang, W. Guan, L. E. Bates, H. T. Stuart, M. Guo, P. Zhang, B. Huang, C. Chen, M. Zhang, J. Chen, M. Min, G. Wu, A. P. Hutchins, and J. C. R. Silva, "In vitro generation of mouse morula-like cells," (in eng), *Developmental cell*, vol. 58, no. 22, pp. 2510-2527.e7, Nov 20 2023, doi: 10.1016/j.devcel.2023.09.013.
- [23] M. Ma, L. Zhang, Z. Liu, Y. Teng, M. Li, X. Peng, and L. An, "Effect of blastocyst development on hatching and embryo implantation," (in eng), *Theriogenology*, vol. 214, pp. 66-72, Jan 15 2024, doi: 10.1016/j.theriogenology.2023.10.011.

- [24] K. M. Schüle, J. Weckerle, S. Probst, A. E. Wehmeyer, L. Zissel, C. M. Schröder, M. Tekman, G. J. Kim, I. M. Schlägl, Sagar, and S. J. Arnold, "Eomes restricts Brachyury functions at the onset of mouse gastrulation," (in eng), *Developmental cell*, vol. 58, no. 18, pp. 1627-1642.e7, Sep 25 2023, doi: 10.1016/j.devcel.2023.07.023.
- [25] S. Y. Oh, S. B. Na, Y. K. Kang, and J. T. Do, "In Vitro Embryogenesis and Gastrulation Using Stem Cells in Mice and Humans," (in eng), *International journal of molecular sciences*, vol. 24, no. 17, Sep 4 2023, doi: 10.3390/ijms241713655.
- [26] A. Aguilera-Castrejon, B. Oldak, T. Shani, N. Ghanem, C. Itzkovich, S. Slomovich, S. Tarazi, J. Bayerl, V. Chugaeva, M. Ayyash, S. Ashouokhi, D. Sheban, N. Livnat, L. Lasman, S. Viukov, M. Zerbib, Y. Addadi, Y. Rais, S. Cheng, Y. Stelzer, H. Keren-Shaul, R. Shlomo, R. Massarwa, N. Novershtern, I. Maza, and J. H. Hanna, "Ex utero mouse embryogenesis from pre-gastrulation to late organogenesis," (in eng), *Nature*, vol. 593, no. 7857, pp. 119-124, May 2021, doi: 10.1038/s41586-021-03416-3.
- [27] F. Qu, W. Li, J. Xu, R. Zhang, J. Ke, X. Ren, X. Meng, L. Qin, J. Zhang, F. Lu, X. Zhou, X. Luo, Z. Zhang, M. Wang, G. Wu, D. Pei, J. Chen, G. Cui, S. Suo, and G. Peng, "Three-dimensional molecular architecture of mouse organogenesis," (in eng), *Nature communications*, vol. 14, no. 1, p. 4599, Jul 31 2023, doi: 10.1038/s41467-023-40155-7.

- [28] S. C. van den Brink and A. van Oudenaarden, "3D gastruloids: a novel frontier in stem cell-based in vitro modeling of mammalian gastrulation," (in eng), *Trends in cell biology*, vol. 31, no. 9, pp. 747-759, Sep 2021, doi: 10.1016/j.tcb.2021.06.007.
- [29] J. Cao, M. Spielmann, X. Qiu, X. Huang, D. M. Ibrahim, A. J. Hill, F. Zhang, S. Mundlos, L. Christiansen, F. J. Steemers, C. Trapnell, and J. Shendure, "The single-cell transcriptional landscape of mammalian organogenesis," (in eng), *Nature*, vol. 566, no. 7745, pp. 496-502, Feb 2019, doi: 10.1038/s41586-019-0969-x.
- [30] E. S. Bardot and A. K. Hadjantonakis, "Mouse gastrulation: Coordination of tissue patterning, specification and diversification of cell fate," (in eng), *Mechanisms of development*, vol. 163, p. 103617, Sep 2020, doi: 10.1016/j.mod.2020.103617.
- [31] L. Xue, J. Y. Cai, J. Ma, Z. Huang, M. X. Guo, L. Z. Fu, Y. B. Shi, and W. X. Li, "Global expression profiling reveals genetic programs underlying the developmental divergence between mouse and human embryogenesis," (in eng), *BMC genomics*, vol. 14, p. 568, Aug 20 2013, doi: 10.1186/1471-2164-14-568.
- [32] S. A. Elmore, R. Z. Cochran, B. Bolon, B. Lubeck, B. Mahler, D. Sabio, and J. M. Ward, "Histology Atlas of the Developing Mouse Placenta," (in eng), *Toxicol Pathol*, vol. 50, no. 1, pp. 60-117, Jan 2022, doi: 10.1177/01926233211042270.
- [33] I. Shioka, R. Morita, R. Yagasaki, D. Wuergeshen, T. Yamashita, H. Fujiwara, and S. Okuda, "Ex vivo SIM-AFM measurements reveal the spatial correlation

- of stiffness and molecular distributions in 3D living tissue," (in eng), *Acta biomaterialia*, vol. 189, pp. 351-365, Nov 2024, doi: 10.1016/j.actbio.2024.09.023.
- [34] A. Battistella, L. Andolfi, M. Zanetti, S. Dal Zilio, M. Stebel, G. Ricci, and M. Lazzarino, "Atomic force spectroscopy-based essay to evaluate oocyte postovulatory aging," (in eng), *Bioengineering & translational medicine*, vol. 7, no. 3, p. e10294, Sep 2022, doi: 10.1002/btm2.10294.
- [35] S. Chakraborty, K. E. Peak, J. P. Gleghorn, T. J. Carroll, and V. D. Varner, "Quantifying Spatial Patterns of Tissue Stiffness Within the Embryonic Mouse Kidney," (in eng), *Methods in molecular biology (Clifton, N.J.)*, vol. 2805, pp. 171-186, 2024, doi: 10.1007/978-1-0716-3854-5_12.
- [36] M. Fluks, A. Rak, and A. Ajduk, "Cytoplasmic movement velocity in unfertilized mouse oocytes: a supportive but not definitive marker of embryo quality," (in eng), *Theriogenology*, vol. 251, p. 117725, Feb 1 2026, doi: 10.1016/j.theriogenology.2025.117725.
- [37] O. J. De La, G. L. Galea, and A. C. Martin, "A lateral tension model for mouse cranial neural tube closure," (in eng), *bioRxiv : the preprint server for biology*, May 15 2025, doi: 10.1101/2025.05.15.654327.
- [38] A. R. Marshall, G. L. Galea, A. J. Copp, and N. D. E. Greene, "The surface ectoderm exhibits spatially heterogenous tension that correlates with YAP localisation during spinal neural tube closure in mouse embryos," (in eng), *Cells*

- & *development*, vol. 174, p. 203840, Jun 2023, doi: 10.1016/j.cdev.2023.203840.
- [39] R. Raghunathan, J. Zhang, C. Wu, J. Rippey, M. Singh, K. V. Larin, and G. Scarcelli, "Evaluating biomechanical properties of murine embryos using Brillouin microscopy and optical coherence tomography," (in eng), *Journal of biomedical optics*, vol. 22, no. 8, pp. 1-6, Aug 2017, doi: 10.1117/1.Jbo.22.8.086013.
- [40] Y. S. Ambekar, M. Singh, A. W. Schill, J. Zhang, C. Zevallos-Delgado, B. Khajavi, S. R. Aglyamov, R. H. Finnell, G. Scarcelli, and K. V. Larin, "Multimodal imaging system combining optical coherence tomography and Brillouin microscopy for neural tube imaging," (in eng), *Opt Lett*, vol. 47, no. 6, pp. 1347-1350, Mar 15 2022, doi: 10.1364/ol.453996.
- [41] Y. S. Ambekar, C. D. Caiaffa, B. J. Wlodarczyk, M. Singh, A. W. Schill, J. W. Steele, J. Zhang, S. R. Aglyamov, G. Scarcelli, R. H. Finnell, and K. V. Larin, "Optical coherence tomography-guided Brillouin microscopy highlights regional tissue stiffness differences during anterior neural tube closure in the *Mthfd11* murine mutant," (in eng), *Development (Cambridge, England)*, vol. 151, no. 10, May 15 2024, doi: 10.1242/dev.202475.
- [42] C. D. Caiaffa, Y. S. Ambekar, M. Singh, Y. L. Lin, B. Wlodarczyk, S. R. Aglyamov, G. Scarcelli, K. V. Larin, and R. H. Finnell, "Disruption of *Fuz* in mouse embryos generates hypoplastic hindbrain development and reduced cranial nerve ganglia," (in eng), *Developmental dynamics : an official*

- publication of the American Association of Anatomists*, vol. 253, no. 9, pp. 846-858, Sep 2024, doi: 10.1002/dvdy.702.
- [43] J. A. Ketterling, O. Aristizábal, B. Y. S. Yiu, D. H. Turnbull, C. K. L. Phoon, A. C. H. Yu, and R. H. Silverman, "High-speed, high-frequency ultrasound, in utero vector-flow imaging of mouse embryos," (in eng), *Scientific reports*, vol. 7, no. 1, p. 16658, Nov 30 2017, doi: 10.1038/s41598-017-16933-x.
 - [44] M. Zhu, K. Zhang, E. C. Thomas, R. Xu, B. Ciruna, S. Hopyan, and Y. Sun, "Tissue stiffness mapping by light sheet elastography," (in eng), *Science advances*, vol. 11, no. 11, p. eadt7274, Mar 14 2025, doi: 10.1126/sciadv.adt7274.
 - [45] M. Singh, F. Zvietcovich, C. Zevallos-Delgado, Y. S. Ambekar, S. R. Aglyamov, and K. V. J. O. Larin, "Whole embryo biomechanics with reverberant optical coherence elastography," vol. 11, no. 5, pp. 686-692, 2024.
 - [46] R. Fuentes and J. Fernández, "Ooplasmic segregation in the zebrafish zygote and early embryo: pattern of ooplasmic movements and transport pathways," (in eng), *Dev Dyn*, vol. 239, no. 8, pp. 2172-89, Aug 2010, doi: 10.1002/dvdy.22349.
 - [47] R. E. Lindeman and F. Pelegri, "Vertebrate maternal-effect genes: Insights into fertilization, early cleavage divisions, and germ cell determinant localization from studies in the zebrafish," (in eng), *Mol Reprod Dev*, vol. 77, no. 4, pp. 299-313, Apr 2010, doi: 10.1002/mrd.21128.

- [48] B. R. Keegan, D. Meyer, and D. Yelon, "Organization of cardiac chamber progenitors in the zebrafish blastula," (in eng), *Development*, vol. 131, no. 13, pp. 3081-91, Jul 2004, doi: 10.1242/dev.01185.
- [49] L. Solnica-Krezel, "Gastrulation in zebrafish -- all just about adhesion?," (in eng), *Curr Opin Genet Dev*, vol. 16, no. 4, pp. 433-41, Aug 2006, doi: 10.1016/j.gde.2006.06.009.
- [50] C. B. Kimmel, W. W. Ballard, S. R. Kimmel, B. Ullmann, and T. F. Schilling, "Stages of embryonic development of the zebrafish," (in eng), *Dev Dyn*, vol. 203, no. 3, pp. 253-310, Jul 1995, doi: 10.1002/aja.1002030302.
- [51] X. Niu, H. Shi, and J. Peng, "The role of mesodermal signals during liver organogenesis in zebrafish," (in eng), *Sci China Life Sci*, vol. 53, no. 4, pp. 455-61, Apr 2010, doi: 10.1007/s11427-010-0078-4.
- [52] S. R. Keenan and P. D. Currie, "The Developmental Phases of Zebrafish Myogenesis," (in eng), *J Dev Biol*, vol. 7, no. 2, Jun 2 2019, doi: 10.3390/jdb7020012.
- [53] D. M. Parichy, M. R. Elizondo, M. G. Mills, T. N. Gordon, and R. E. Engeszer, "Normal table of postembryonic zebrafish development: staging by externally visible anatomy of the living fish," (in eng), *Dev Dyn*, vol. 238, no. 12, pp. 2975-3015, Dec 2009, doi: 10.1002/dvdy.22113.
- [54] M. J. Hosen, O. M. Vanakker, A. Willaert, A. Huysseune, P. Coucke, and A. De Paepe, "Zebrafish models for ectopic mineralization disorders: practical issues

- from morpholino design to post-injection observations," (in eng), *Front Genet*, vol. 4, p. 74, 2013, doi: 10.3389/fgene.2013.00074.
- [55] M. Marsal, A. Hernández-Vega, and E. Martin-Blanco, "Contractility, differential tension and membrane removal lead zebrafish epiboly biomechanics," (in eng), *Cell Cycle*, vol. 16, no. 14, pp. 1328-1335, Jul 18 2017, doi: 10.1080/15384101.2017.1327489.
- [56] D. Bornhorst, P. Xia, H. Nakajima, C. Dingare, W. Herzog, V. Lecaudey, N. Mochizuki, C. P. Heisenberg, D. Yelon, and S. Abdelilah-Seyfried, "Biomechanical signaling within the developing zebrafish heart attunes endocardial growth to myocardial chamber dimensions," (in eng), *Nat Commun*, vol. 10, no. 1, p. 4113, Sep 11 2019, doi: 10.1038/s41467-019-12068-x.
- [57] A. Paolini and S. Abdelilah-Seyfried, "The mechanobiology of zebrafish cardiac valve leaflet formation," (in eng), *Curr Opin Cell Biol*, vol. 55, pp. 52-58, Dec 2018, doi: 10.1016/j.ceb.2018.05.007.
- [58] C. J. Voesenek, F. T. Muijres, and J. L. van Leeuwen, "Biomechanics of swimming in developing larval fish," (in eng), *J Exp Biol*, vol. 221, no. Pt 1, Jan 11 2018, doi: 10.1242/jeb.149583.
- [59] P. Drapeau, L. Saint-Amant, R. R. Buss, M. Chong, J. R. McDearnid, and E. Brustein, "Development of the locomotor network in zebrafish," (in eng), *Prog Neurobiol*, vol. 68, no. 2, pp. 85-111, Oct 2002, doi: 10.1016/s0301-0082(02)00075-8.

- [60] T. Sugioka, M. Tanimoto, and S. I. Higashijima, "Biomechanics and neural circuits for vestibular-induced fine postural control in larval zebrafish," (in eng), *Nat Commun*, vol. 14, no. 1, p. 1217, Mar 10 2023, doi: 10.1038/s41467-023-36682-y.
- [61] D. H. Kim, C. N. Hwang, Y. Sun, S. H. Lee, B. Kim, and B. J. Nelson, "Mechanical analysis of chorion softening in prehatching stages of zebrafish embryos," (in eng), *IEEE Trans Nanobioscience*, vol. 5, no. 2, pp. 89-94, Jun 2006, doi: 10.1109/tnb.2006.875054.
- [62] Z. Lu, P. C. Chen, H. Luo, J. Nam, R. Ge, and W. Lin, "Models of maximum stress and strain of zebrafish embryos under indentation," (in eng), *J Biomech*, vol. 42, no. 5, pp. 620-5, Mar 26 2009, doi: 10.1016/j.jbiomech.2008.12.017.
- [63] C. Bevilacqua, H. Sánchez-Iranzo, D. Richter, A. Diz-Muñoz, and R. Prevedel, "Imaging mechanical properties of sub-micron ECM in live zebrafish using Brillouin microscopy," (in eng), *Biomed Opt Express*, vol. 10, no. 3, pp. 1420-1431, Mar 1 2019, doi: 10.1364/boe.10.001420.
- [64] R. Schlüßler, S. Möllmert, S. Abuhattum, G. Cojoc, P. Müller, K. Kim, C. Möckel, C. Zimmermann, J. Czarske, and J. Guck, "Mechanical Mapping of Spinal Cord Growth and Repair in Living Zebrafish Larvae by Brillouin Imaging," (in eng), *Biophys J*, vol. 115, no. 5, pp. 911-923, Sep 4 2018, doi: 10.1016/j.bpj.2018.07.027.

- [65] H. Sánchez-Iranzo, C. Bevilacqua, A. Diz-Muñoz, and R. Prevedel, "A 3D Brillouin microscopy dataset of the in-vivo zebrafish eye," (in eng), *Data Brief*, vol. 30, p. 105427, Jun 2020, doi: 10.1016/j.dib.2020.105427.
- [66] C. Zevallos-Delgado, T. T. Mekonnen, Y. S. Ambekar, M. Singh, J. D. Wythe, S. R. Aglyamov, and K. V. Larin, "Assessment of the biomechanical changes on zebrafish larvae eye development caused by phenylthiourea (PTU) treatment using reverberant optical coherence elastography (Rev-OCE)," in *Optical Elastography and Tissue Biomechanics XI*, 2024, vol. 12844: SPIE, pp. 18-21.
- [67] B. S. Garra, "Elastography: history, principles, and technique comparison," (in eng), *Abdom Imaging*, vol. 40, no. 4, pp. 680-97, Apr 2015, doi: 10.1007/s00261-014-0305-8.
- [68] J. Ophir, I. Céspedes, H. Ponnekanti, Y. Yazdi, and X. Li, "Elastography: a quantitative method for imaging the elasticity of biological tissues," (in eng), *Ultrason Imaging*, vol. 13, no. 2, pp. 111-34, Apr 1991, doi: 10.1177/016173469101300201.
- [69] M. Tristam, D. C. Barbosa, D. O. Cosgrove, D. K. Nassiri, J. C. Bamber, and C. R. Hill, "Ultrasonic study of in vivo kinetic characteristics of human tissues," (in eng), *Ultrasound Med Biol*, vol. 12, no. 12, pp. 927-37, Dec 1986, doi: 10.1016/0301-5629(86)90061-x.
- [70] K. J. Parker, S. R. Huang, R. A. Musulin, and R. M. Lerner, "Tissue response to mechanical vibrations for "sonoelasticity imaging"," (in eng), *Ultrasound Med Biol*, vol. 16, no. 3, pp. 241-6, 1990, doi: 10.1016/0301-5629(90)90003-u.

- [71] Y. Yamakoshi, J. Sato, and T. Sato, "Ultrasonic imaging of internal vibration of soft tissue under forced vibration," (in eng), *IEEE Trans Ultrason Ferroelectr Freq Control*, vol. 37, no. 2, pp. 45-53, 1990, doi: 10.1109/58.46969.
- [72] G. Scarcelli and S. H. Yun, "Confocal Brillouin microscopy for three-dimensional mechanical imaging," (in eng), *Nat Photonics*, vol. 2, pp. 39-43, Dec 9 2007, doi: 10.1038/nphoton.2007.250.
- [73] J. Schumacher, C. Zevallos-Delgado, L. Rohman, A. W. Schill, M. Singh, H. Zhang, M. Ruggeri, J. M. Parel, F. Manns, K. V. Larin, and G. Scarcelli, "Clinical multimodal Brillouin microscopy-optical coherence elastography system for lens biomechanics," (in eng), *J Biomed Opt*, vol. 30, no. 12, p. 124511, Dec 2025, doi: 10.1117/1.Jbo.30.12.124511.
- [74] X. W. Cui, K. N. Li, A. J. Yi, B. Wang, Q. Wei, G. G. Wu, and C. F. Dietrich, "Ultrasound elastography," (in eng), *Endosc Ultrasound*, vol. 11, no. 4, pp. 252-274, Jul-Aug 2022, doi: 10.4103/eus-d-21-00151.
- [75] R. M. S. Sigrist, J. Liao, A. E. Kaffas, M. C. Chammas, and J. K. Willmann, "Ultrasound Elastography: Review of Techniques and Clinical Applications," (in eng), *Theranostics*, vol. 7, no. 5, pp. 1303-1329, 2017, doi: 10.7150/thno.18650.
- [76] Y. K. Mariappan, K. J. Glaser, and R. L. Ehman, "Magnetic resonance elastography: a review," (in eng), *Clin Anat*, vol. 23, no. 5, pp. 497-511, Jul 2010, doi: 10.1002/ca.21006.

- [77] A. Manduca, T. E. Oliphant, M. A. Dresner, J. L. Mahowald, S. A. Kruse, E. Amromin, J. P. Felmlee, J. F. Greenleaf, and R. L. Ehman, "Magnetic resonance elastography: non-invasive mapping of tissue elasticity," (in eng), *Med Image Anal*, vol. 5, no. 4, pp. 237-54, Dec 2001, doi: 10.1016/s1361-8415(00)00039-6.
- [78] H. Haga, S. Sasaki, K. Kawabata, E. Ito, T. Ushiki, and T. Sambongi, "Elasticity mapping of living fibroblasts by AFM and immunofluorescence observation of the cytoskeleton," (in eng), *Ultramicroscopy*, vol. 82, no. 1-4, pp. 253-8, Feb 2000, doi: 10.1016/s0304-3991(99)00157-6.
- [79] A. Singh, F. Pati, and R. John, "Quantifying viscosity and elasticity using holographic imaging by Rayleigh wave dispersion," (in eng), *Opt Lett*, vol. 47, no. 9, pp. 2214-2217, May 1 2022, doi: 10.1364/ol.451464.
- [80] K. D. Mohan and A. L. Oldenburg, "Elastography of soft materials and tissues by holographic imaging of surface acoustic waves," (in eng), *Opt Express*, vol. 20, no. 17, pp. 18887-97, Aug 13 2012, doi: 10.1364/oe.20.018887.
- [81] J. Schmitt, "OCT elastography: imaging microscopic deformation and strain of tissue," (in eng), *Opt Express*, vol. 3, no. 6, pp. 199-211, Sep 14 1998, doi: 10.1364/oe.3.000199.
- [82] D. Huang, E. A. Swanson, C. P. Lin, J. S. Schuman, W. G. Stinson, W. Chang, M. R. Hee, T. Flotte, K. Gregory, C. A. Puliafito, and et al., "Optical coherence tomography," (in eng), *Science*, vol. 254, no. 5035, pp. 1178-81, Nov 22 1991, doi: 10.1126/science.1957169.

- [83] D. P. Popescu, L. P. Choo-Smith, C. Flueraru, Y. Mao, S. Chang, J. Disano, S. Sherif, and M. G. Sowa, "Optical coherence tomography: fundamental principles, instrumental designs and biomedical applications," (in eng), *Biophys Rev*, vol. 3, no. 3, p. 155, Sep 2011, doi: 10.1007/s12551-011-0054-7.
- [84] M. Zeppieri, S. Marsili, E. S. Enaholo, A. O. Shuaibu, N. Uwagboe, C. Salati, L. Spadea, and M. Musa, "Optical Coherence Tomography (OCT): A Brief Look at the Uses and Technological Evolution of Ophthalmology," (in eng), *Medicina (Kaunas)*, vol. 59, no. 12, Dec 3 2023, doi: 10.3390/medicina59122114.
- [85] J. F. Bille, "High resolution imaging in microscopy and ophthalmology: new frontiers in biomedical optics," 2019.
- [86] J. Cheng, D. Tao, Y. Quan, D. W. Wong, G. C. Cheung, M. Akiba, and J. Liu, "Speckle Reduction in 3D Optical Coherence Tomography of Retina by A-Scan Reconstruction," (in eng), *IEEE Trans Med Imaging*, vol. 35, no. 10, pp. 2270-2279, Oct 2016, doi: 10.1109/tmi.2016.2556080.
- [87] E. Bousi, I. Charalambous, and C. Pitris, "Optical coherence tomography axial resolution improvement by step-frequency encoding," (in eng), *Opt Express*, vol. 18, no. 11, pp. 11877-90, May 24 2010, doi: 10.1364/oe.18.011877.
- [88] J. Welzel, "Optical coherence tomography in dermatology: a review," (in eng), *Skin Res Technol*, vol. 7, no. 1, pp. 1-9, Feb 2001, doi: 10.1034/j.1600-0846.2001.007001001.x.

- [89] V. R. Shidlovski, "Superluminescent diode light sources for OCT," in *Optical Coherence Tomography: Technology and Applications*: Springer, 2008, pp. 281-299.
- [90] K. S. Park, E. Park, H. Lee, H. J. Lee, S. W. Lee, and T. J. Eom, "Phase stable swept-source optical coherence tomography with active mode-locking laser for contrast enhancements of retinal angiography," (in eng), *Sci Rep*, vol. 11, no. 1, p. 16636, Aug 17 2021, doi: 10.1038/s41598-021-95982-9.
- [91] K. Irsch, "Optical principles of OCT," in *Albert and Jakobiec's Principles and Practice of Ophthalmology*: Springer, 2022, pp. 1083-1096.
- [92] R. Leitgeb, C. Hitzenberger, and A. Fercher, "Performance of fourier domain vs. time domain optical coherence tomography," (in eng), *Opt Express*, vol. 11, no. 8, pp. 889-94, Apr 21 2003, doi: 10.1364/oe.11.000889.
- [93] J. M. Schmitt, A. Knüttel, M. Yadlowsky, and M. A. Eckhaus, "Optical-coherence tomography of a dense tissue: statistics of attenuation and backscattering," (in eng), *Phys Med Biol*, vol. 39, no. 10, pp. 1705-20, Oct 1994, doi: 10.1088/0031-9155/39/10/013.
- [94] Y. Pan, R. Birngruber, J. Rosperich, and R. Engelhardt, "Low-coherence optical tomography in turbid tissue: theoretical analysis," (in eng), *Appl Opt*, vol. 34, no. 28, pp. 6564-74, Oct 1 1995, doi: 10.1364/ao.34.006564.
- [95] A. G. Podoleanu and D. A. Jackson, "Noise analysis of a combined optical coherence tomograph and a confocal scanning ophthalmoscope," (in eng), *Appl Opt*, vol. 38, no. 10, pp. 2116-27, Apr 1 1999, doi: 10.1364/ao.38.002116.

- [96] S. R. Chinn, E. A. Swanson, and J. G. Fujimoto, "Optical coherence tomography using a frequency-tunable optical source," (in eng), *Opt Lett*, vol. 22, no. 5, pp. 340-2, Mar 1 1997, doi: 10.1364/ol.22.000340.
- [97] B. Golubovic, B. E. Bouma, G. J. Tearney, and J. G. Fujimoto, "Optical frequency-domain reflectometry using rapid wavelength tuning of a Cr4+:forsterite laser," (in eng), *Opt Lett*, vol. 22, no. 22, pp. 1704-6, Nov 15 1997, doi: 10.1364/ol.22.001704.
- [98] H. Hiratsuka, E. Kido, and T. Yoshimura, "Simultaneous measurements of three-dimensional reflectivity distributions in scattering media based on optical frequency-domain reflectometry," (in eng), *Opt Lett*, vol. 23, no. 18, pp. 1420-2, Sep 15 1998, doi: 10.1364/ol.23.001420.
- [99] M. Choma, M. Sarunic, C. Yang, and J. Izatt, "Sensitivity advantage of swept source and Fourier domain optical coherence tomography," (in eng), *Opt Express*, vol. 11, no. 18, pp. 2183-9, Sep 8 2003, doi: 10.1364/oe.11.002183.
- [100] A. F. Fercher, C. K. Hitzenberger, G. Kamp, and S. Y. El-Zaiat, "Measurement of intraocular distances by backscattering spectral interferometry," *Optics communications*, vol. 117, no. 1-2, pp. 43-48, 1995.
- [101] F. Lexer, C. K. Hitzenberger, A. F. Fercher, and M. Kulhavy, "Wavelength-tuning interferometry of intraocular distances," (in eng), *Appl Opt*, vol. 36, no. 25, pp. 6548-53, Sep 1 1997, doi: 10.1364/ao.36.006548.
- [102] J. F. de Boer, R. Leitgeb, and M. Wojtkowski, "Twenty-five years of optical coherence tomography: the paradigm shift in sensitivity and speed provided by

- Fourier domain OCT [Invited]," (in eng), *Biomed Opt Express*, vol. 8, no. 7, pp. 3248-3280, Jul 1 2017, doi: 10.1364/boe.8.003248.
- [103] Z. Yaqoob, J. Wu, and C. Yang, "Spectral domain optical coherence tomography: a better OCT imaging strategy," (in eng), *Biotechniques*, vol. 39, no. 6 Suppl, pp. S6-13, Dec 2005, doi: 10.2144/000112090.
- [104] D. F. Kiernan, W. F. Mieler, and S. M. Hariprasad, "Spectral-domain optical coherence tomography: a comparison of modern high-resolution retinal imaging systems," (in eng), *Am J Ophthalmol*, vol. 149, no. 1, pp. 18-31, Jan 2010, doi: 10.1016/j.ajo.2009.08.037.
- [105] F. Xia and R. Hua, "The Latest Updates in Swept-Source Optical Coherence Tomography Angiography," (in eng), *Diagnostics (Basel)*, vol. 14, no. 1, Dec 25 2023, doi: 10.3390/diagnostics14010047.
- [106] S. Aumann, S. Donner, J. Fischer, and F. Müller, "Optical Coherence Tomography (OCT): Principle and Technical Realization," in *High Resolution Imaging in Microscopy and Ophthalmology: New Frontiers in Biomedical Optics*, J. F. Bille Ed. Cham (CH): Springer
Copyright 2019, The Author(s). 2019, pp. 59-85.
- [107] Y. Jia, T. T. Hormel, T. S. Hwang, A. L. Wu, G. B. Liang, Y. Guo, X. Wei, S. Ni, Y. Jian, J. P. Campbell, S. T. Bailey, J. C. Morrison, and D. Huang, "Widefield OCT angiography," (in eng), *Prog Retin Eye Res*, vol. 107, p. 101378, Jul 2025, doi: 10.1016/j.preteyeres.2025.101378.

- [108] R. S. Jonnal, O. P. Kocaoglu, R. J. Zawadzki, Z. Liu, D. T. Miller, and J. S. Werner, "A Review of Adaptive Optics Optical Coherence Tomography: Technical Advances, Scientific Applications, and the Future," (in eng), *Invest Ophthalmol Vis Sci*, vol. 57, no. 9, pp. Oct51-68, Jul 1 2016, doi: 10.1167/iovs.16-19103.
- [109] M.-Y. Fu, Z.-H. Yin, X.-Y. Yao, J. Xu, Y. Liu, Y. Dong, and Y.-C. Shen, "The progress of optical coherence tomography in industry applications," *Advanced Devices & Instrumentation*, vol. 5, p. 0053, 2024.
- [110] M. L. Yang, C. W. Lu, I. J. Hsu, and C. Yang, "The use of optical coherence tomography for monitoring the subsurface morphologies of archaic jades," *Archaeometry*, vol. 46, no. 2, pp. 171-182, 2004.
- [111] H. Liang, M. Cid, R. Cucu, G. Dobre, A. Podoleanu, J. Pedro, and D. Saunders, "En-face optical coherence tomography - a novel application of non-invasive imaging to art conservation," (in eng), *Opt Express*, vol. 13, no. 16, pp. 6133-44, Aug 8 2005, doi: 10.1364/opex.13.006133.
- [112] J. G. Fujimoto, W. Drexler, J. S. Schuman, and C. K. Hitzenberger, "Optical Coherence Tomography (OCT) in ophthalmology: introduction," (in eng), *Opt Express*, vol. 17, no. 5, pp. 3978-9, Mar 2 2009, doi: 10.1364/oe.17.003978.
- [113] T. Yonetsu, B. E. Bouma, K. Kato, J. G. Fujimoto, and I. K. Jang, "Optical coherence tomography— 15 years in cardiology," (in eng), *Circ J*, vol. 77, no. 8, pp. 1933-40, 2013, doi: 10.1253/circj.cj-13-0643.1.

- [114] L. Yang, Y. Chen, S. Ling, J. Wang, G. Wang, B. Zhang, H. Zhao, Q. Zhao, and J. Mao, "Research progress on the application of optical coherence tomography in the field of oncology," (in eng), *Front Oncol*, vol. 12, p. 953934, 2022, doi: 10.3389/fonc.2022.953934.
- [115] J. Olsen, J. Holmes, and G. B. Jemec, "Advances in optical coherence tomography in dermatology-a review," (in eng), *J Biomed Opt*, vol. 23, no. 4, pp. 1-10, Apr 2018, doi: 10.1117/1.Jbo.23.4.040901.
- [116] A. Al-Ani, É. Benard-Seguin, and F. Costello, "Optical coherence tomography: implications for neurology," (in eng), *Curr Opin Neurol*, vol. 38, no. 1, pp. 62-70, Feb 1 2025, doi: 10.1097/wco.0000000000001340.
- [117] T. H. Tsai, C. L. Leggett, A. J. Trindade, A. Sethi, A. F. Swager, V. Joshi, J. J. Bergman, H. Mashimo, N. S. Nishioka, and E. Namati, "Optical coherence tomography in gastroenterology: a review and future outlook," (in eng), *J Biomed Opt*, vol. 22, no. 12, pp. 1-17, Dec 2017, doi: 10.1117/1.Jbo.22.12.121716.
- [118] S. A. Boppart, M. E. Brezinski, and J. G. Fujimoto, "Optical coherence tomography imaging in developmental biology," (in eng), *Methods Mol Biol*, vol. 135, pp. 217-33, 2000, doi: 10.1385/1-59259-685-1:217.
- [119] D. M. Scully and I. V. Larina, "Mouse embryo phenotyping with optical coherence tomography," (in eng), *Front Cell Dev Biol*, vol. 10, p. 1000237, 2022, doi: 10.3389/fcell.2022.1000237.

- [120] S. Wang, M. D. Garcia, A. L. Lopez, 3rd, P. A. Overbeek, K. V. Larin, and I. V. Larina, "Dynamic imaging and quantitative analysis of cranial neural tube closure in the mouse embryo using optical coherence tomography," (in eng), *Biomed Opt Express*, vol. 8, no. 1, pp. 407-419, Jan 1 2017, doi: 10.1364/boe.8.000407.
- [121] S. Wang and I. Larina, "Live mechanistic assessment of localized cardiac pumping in mammalian tubular embryonic heart," (in eng), *J Biomed Opt*, vol. 25, no. 8, pp. 1-19, Aug 2020, doi: 10.1117/1.Jbo.25.8.086001.
- [122] A. L. Lopez, 3rd, S. Wang, and I. V. Larina, "Embryonic Mouse Cardiodynamic OCT Imaging," (in eng), *J Cardiovasc Dev Dis*, vol. 7, no. 4, Oct 4 2020, doi: 10.3390/jcdd7040042.
- [123] S. Wang and I. V. Larina, "In vivo dynamic 3D imaging of oocytes and embryos in the mouse oviduct," (in eng), *Cell Rep*, vol. 36, no. 2, p. 109382, Jul 13 2021, doi: 10.1016/j.celrep.2021.109382.
- [124] S. Rodimova, E. Gubarkova, N. Bobrov, I. Shchekhin, V. Kozlova, N. Zolotova, A. Potapov, E. Kiseleva, G. Gelikonov, N. Gladkova, V. Zagainov, E. Zagaynova, and D. Kuznetsova, "Optical Coherence Tomography Angiography, Elastography, and Attenuation Imaging for Evaluation of Liver Regeneration," (in eng), *Diagnostics (Basel)*, vol. 15, no. 8, Apr 11 2025, doi: 10.3390/diagnostics15080977.
- [125] A. H. Kashani, C. L. Chen, J. K. Gahm, F. Zheng, G. M. Richter, P. J. Rosenfeld, Y. Shi, and R. K. Wang, "Optical coherence tomography angiography: A

- comprehensive review of current methods and clinical applications," (in eng), *Prog Retin Eye Res*, vol. 60, pp. 66-100, Sep 2017, doi: 10.1016/j.preteyeres.2017.07.002.
- [126] D. P. Davé, T. Akkin, and T. E. Milner, "Polarization-maintaining fiber-based optical low-coherence reflectometer for characterization and ranging of birefringence," (in eng), *Opt Lett*, vol. 28, no. 19, pp. 1775-7, Oct 1 2003, doi: 10.1364/ol.28.001775.
- [127] U. Morgner, W. Drexler, F. X. Kärtner, X. D. Li, C. Pitris, E. P. Ippen, and J. G. Fujimoto, "Spectroscopic optical coherence tomography," (in eng), *Opt Lett*, vol. 25, no. 2, pp. 111-3, Jan 15 2000, doi: 10.1364/ol.25.000111.
- [128] M. Singh, J. Li, S. Vantipalli, S. Wang, Z. Han, A. Nair, S. R. Aglyamov, M. D. Twa, and K. V. Larin, "Noncontact Elastic Wave Imaging Optical Coherence Elastography for Evaluating Changes in Corneal Elasticity Due to Crosslinking," (in eng), *IEEE J Sel Top Quantum Electron*, vol. 22, no. 3, May-Jun 2016, doi: 10.1109/jstqe.2015.2510293.
- [129] K. M. Kennedy, R. A. McLaughlin, B. F. Kennedy, A. Tien, B. Latham, C. M. Saunders, and D. D. Sampson, "Needle optical coherence elastography for the measurement of microscale mechanical contrast deep within human breast tissues," (in eng), *J Biomed Opt*, vol. 18, no. 12, p. 121510, Dec 2013, doi: 10.1117/1.Jbo.18.12.121510.

- [130] K. V. Larin and D. D. Sampson, "Optical coherence elastography - OCT at work in tissue biomechanics [Invited]," (in eng), *Biomed Opt Express*, vol. 8, no. 2, pp. 1172-1202, Feb 1 2017, doi: 10.1364/boe.8.001172.
- [131] K. M. Kennedy, L. Chin, R. A. McLaughlin, B. Latham, C. M. Saunders, D. D. Sampson, and B. F. Kennedy, "Quantitative micro-elastography: imaging of tissue elasticity using compression optical coherence elastography," (in eng), *Sci Rep*, vol. 5, p. 15538, Oct 27 2015, doi: 10.1038/srep15538.
- [132] B. F. Kennedy, K. M. Kennedy, and D. D. Sampson, "A review of optical coherence elastography: fundamentals, techniques and prospects," *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 20, no. 2, pp. 272-288, 2013.
- [133] M. A. Kirby, I. Pelivanov, S. Song, Ł. Ambrozinski, S. J. Yoon, L. Gao, D. Li, T. T. Shen, R. K. Wang, and M. O'Donnell, "Optical coherence elastography in ophthalmology," (in eng), *J Biomed Opt*, vol. 22, no. 12, pp. 1-28, Dec 2017, doi: 10.1117/1.Jbo.22.12.121720.
- [134] V. Y. Zaitsev, A. L. Matveyev, L. A. Matveev, A. A. Sovetsky, M. S. Hepburn, A. Mowla, and B. F. Kennedy, "Strain and elasticity imaging in compression optical coherence elastography: The two-decade perspective and recent advances," (in eng), *J Biophotonics*, vol. 14, no. 2, p. e202000257, Feb 2021, doi: 10.1002/jbio.202000257.
- [135] B. F. Kennedy, R. A. McLaughlin, K. M. Kennedy, L. Chin, A. Curatolo, A. Tien, B. Latham, C. M. Saunders, and D. D. Sampson, "Optical coherence

- micro-elastography: mechanical-contrast imaging of tissue microstructure," (in eng), *Biomed Opt Express*, vol. 5, no. 7, pp. 2113-24, Jul 1 2014, doi: 10.1364/boe.5.002113.
- [136] M. S. Hepburn, P. Wijesinghe, L. Chin, and B. F. Kennedy, "Analysis of spatial resolution in phase-sensitive compression optical coherence elastography," (in eng), *Biomed Opt Express*, vol. 10, no. 3, pp. 1496-1513, Mar 1 2019, doi: 10.1364/boe.10.001496.
- [137] M. Levy and H. E. Bass, "Handbook of elastic properties of solids, liquids, and gases," ed: Acoustical Society of America, 2004.
- [138] K. F. Graff, *Wave motion in elastic solids*. Courier Corporation, 2012.
- [139] E. L. Carstensen and K. J. Parker, "Oestreicher and elastography," (in eng), *J Acoust Soc Am*, vol. 138, no. 4, pp. 2317-25, Oct 2015, doi: 10.1121/1.4930953.
- [140] I. A. Viktorov, "Rayleigh and Lamb waves," *Rayleigh and Lamb waves*, p. 113, 1967.
- [141] F. Zvietcovich and K. V. Larin, "Wave-based optical coherence elastography: The 10-year perspective," (in eng), *Prog Biomed Eng (Bristol)*, vol. 4, no. 1, Jan 2022, doi: 10.1088/2516-1091/ac4512.
- [142] M. Singh, F. Zvietcovich, and K. V. Larin, "Introduction to optical coherence elastography: tutorial," (in eng), *J Opt Soc Am A Opt Image Sci Vis*, vol. 39, no. 3, pp. 418-430, Mar 1 2022, doi: 10.1364/josaa.444808.
- [143] W. Drexler, M. Liu, A. Kumar, T. Kamali, A. Unterhuber, and R. A. Leitgeb, "Optical coherence tomography today: speed, contrast, and multimodality," (in

- eng), *J Biomed Opt*, vol. 19, no. 7, p. 071412, 2014, doi: 10.1117/1.Jbo.19.7.071412.
- [144] S. Wang and K. V. Larin, "Shear wave imaging optical coherence tomography (SWI-OCT) for ocular tissue biomechanics," (in eng), *Opt Lett*, vol. 39, no. 1, pp. 41-4, Jan 1 2014, doi: 10.1364/ol.39.000041.
- [145] S. Song, Z. Huang, and R. K. Wang, "Tracking mechanical wave propagation within tissue using phase-sensitive optical coherence tomography: motion artifact and its compensation," (in eng), *J Biomed Opt*, vol. 18, no. 12, p. 121505, Dec 2013, doi: 10.1117/1.Jbo.18.12.121505.
- [146] S. Song, W. Wei, B. Y. Hsieh, I. Pelivanov, T. T. Shen, M. O'Donnell, and R. K. Wang, "Strategies to improve phase-stability of ultrafast swept source optical coherence tomography for single shot imaging of transient mechanical waves at 16 kHz frame rate," (in eng), *Appl Phys Lett*, vol. 108, no. 19, p. 191104, May 9 2016, doi: 10.1063/1.4949469.
- [147] Ł. Ambroziński, S. Song, S. J. Yoon, I. Pelivanov, D. Li, L. Gao, T. T. Shen, R. K. Wang, and M. O'Donnell, "Acoustic micro-tapping for non-contact 4D imaging of tissue elasticity," (in eng), *Sci Rep*, vol. 6, p. 38967, Dec 23 2016, doi: 10.1038/srep38967.
- [148] G. Schmidt, B. E. Bouma, and N. Uribe-Patarroyo, "Asynchronous, semi-reverberant elastography," (in eng), *Optica*, vol. 11, no. 9, pp. 1285-1294, Sep 20 2024, doi: 10.1364/optica.528507.

- [149] F. Zvietcovich, A. Nair, Y. S. Ambekar, M. Singh, S. R. Aglyamov, M. D. Twa, and K. V. Larin, "Confocal air-coupled ultrasonic optical coherence elastography probe for quantitative biomechanics," (in eng), *Opt Lett*, vol. 45, no. 23, pp. 6567-6570, Dec 1 2020, doi: 10.1364/ol.410593.
- [150] F. Zvietcovich, A. Nair, M. Singh, S. R. Aglyamov, M. D. Twa, and K. V. Larin, "In vivo assessment of corneal biomechanics under a localized cross-linking treatment using confocal air-coupled optical coherence elastography," (in eng), *Biomed Opt Express*, vol. 13, no. 5, pp. 2644-2654, May 1 2022, doi: 10.1364/boe.456186.
- [151] C. Duvvuri, M. Singh, G. Lan, S. R. Aglyamov, K. V. Larin, and M. D. Twa, "Determinants of Human Corneal Mechanical Wave Dispersion for In Vivo Optical Coherence Elastography," (in eng), *Transl Vis Sci Technol*, vol. 14, no. 1, p. 26, Jan 2 2025, doi: 10.1167/tvst.14.1.26.
- [152] A. Ramier, B. Tavakol, and S. H. Yun, "Measuring mechanical wave speed, dispersion, and viscoelastic modulus of the cornea using optical coherence elastography," (in eng), *Opt Express*, vol. 27, no. 12, pp. 16635-16649, Jun 10 2019, doi: 10.1364/oe.27.016635.
- [153] P. Meemon, J. Yao, Y. J. Chu, F. Zvietcovich, K. J. Parker, and J. P. Rolland, "Crawling wave optical coherence elastography," (in eng), *Opt Lett*, vol. 41, no. 5, pp. 847-50, Mar 1 2016, doi: 10.1364/ol.41.000847.

- [154] K. J. Parker, J. Ormachea, F. Zvietcovich, and B. Castaneda, "Reverberant shear wave fields and estimation of tissue properties," (in eng), *Phys Med Biol*, vol. 62, no. 3, pp. 1046-1061, Feb 7 2017, doi: 10.1088/1361-6560/aa5201.
- [155] F. Zvietcovich, P. Pongchalee, P. Meemon, J. P. Rolland, and K. J. Parker, "Reverberant 3D optical coherence elastography maps the elasticity of individual corneal layers," (in eng), *Nat Commun*, vol. 10, no. 1, p. 4895, Oct 25 2019, doi: 10.1038/s41467-019-12803-4.
- [156] J. Ormachea, B. Castaneda, and K. J. Parker, "Shear Wave Speed Estimation Using Reverberant Shear Wave Fields: Implementation and Feasibility Studies," (in eng), *Ultrasound Med Biol*, vol. 44, no. 5, pp. 963-977, May 2018, doi: 10.1016/j.ultrasmedbio.2018.01.011.
- [157] J. Ormachea, K. J. Parker, and R. G. Barr, "An initial study of complete 2D shear wave dispersion images using a reverberant shear wave field," (in eng), *Phys Med Biol*, vol. 64, no. 14, p. 145009, Jul 16 2019, doi: 10.1088/1361-6560/ab2778.
- [158] J. Ormachea, R. G. Barr, and K. J. Parker, "2-D Shear wave dispersion images using the reverberant shear wave field approach: Application in tissues exhibiting power law response," in *2019 IEEE international ultrasonics symposium (IUS)*, 2019: IEEE, pp. 205-208.
- [159] G. Flores, J. Ormachea, S. E. Romero, F. Zvietcovich, K. J. Parker, and B. Castaneda, "Experimental study to evaluate the generation of reverberant shear

- wave fields (R-SWF) in homogenous media," in *2020 IEEE international ultrasonics symposium (IUS)*, 2020: IEEE, pp. 1-4.
- [160] J. Ormachea and K. J. Parker, "Reverberant shear wave phase gradients for elastography," (in eng), *Phys Med Biol*, vol. 66, no. 17, Aug 23 2021, doi: 10.1088/1361-6560/ac1b37.
 - [161] S. E. Romero, R. Naemi, G. Flores, D. Allan, J. Ormachea, E. Gutierrez, F. L. Casado, and B. Castaneda, "Plantar Soft Tissue Characterization Using Reverberant Shear Wave Elastography: A Proof-of-Concept Study," (in eng), *Ultrasound Med Biol*, vol. 48, no. 1, pp. 35-46, Jan 2022, doi: 10.1016/j.ultrasmedbio.2021.09.011.
 - [162] K. J. Parker, J. Ormachea, and Z. Hah, "Group versus Phase Velocity of Shear Waves in Soft Tissues," (in eng), *Ultrason Imaging*, vol. 40, no. 6, pp. 343-356, Nov 2018, doi: 10.1177/0161734618796217.
 - [163] S. S. Poul, J. Ormachea, G. R. Ge, and K. J. Parker, "Comprehensive experimental assessments of rheological models' performance in elastography of soft tissues," (in eng), *Acta Biomater*, vol. 146, pp. 259-273, Jul 1 2022, doi: 10.1016/j.actbio.2022.04.047.
 - [164] H. Asemani, I. E. Kabir, J. Ormachea, M. M. Doyley, J. P. Rolland, and K. J. Parker, "Angular Integral Autocorrelation for Speed Estimation in Shear-Wave Elastography," in *Acoustics*, 2024, vol. 6, no. 2: MDPI, pp. 413-438.
 - [165] R. G. Gary, F. Zvietcovich, J. P. Rolland, H. Mestre, M. Giannetto, M. Nedergaard, and K. J. Parker, "A preliminary study on using reverberant shear

- wave fields in optical coherence elastography to examine mice brain ex vivo," in *Optical Elastography and Tissue Biomechanics VI*, 2019, vol. 10880: SPIE, pp. 55-61.
- [166] L. A. Aleman-Castañeda, F. Zvietcovich, and K. J. Parker, "Reverberant elastography for the elastic characterization of anisotropic tissues," *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 27, no. 4, pp. 1-12, 2021.
- [167] H. Xu, F. Yang, T. Liang, and Z. P. Luo, "Noncontact elasticity measurement of hydrogels in a culture dish using reverberant optical coherence elastography," (in eng), *J Biomech*, vol. 169, p. 112154, May 2024, doi: 10.1016/j.jbiomech.2024.112154.
- [168] H. Xu, F. Yang, T. Liang, W. Zhang, J. Mo, and Z. Luo, "Reverberant optical coherence elastography using 3D-printed randomly distributed scatterers: elasticity mapping of hydrogels in culture dishes," (in eng), *J Biomed Opt*, vol. 30, no. 12, p. 124507, Dec 2025, doi: 10.1117/1.Jbo.30.12.124507.
- [169] N. Kunanta, F. Zvietcovich, K. J. Parker, J. P. Rolland, and P. Meemon, "Performance of Shear Wave Speed Measurements by Using Reverberant Optical Coherence Elastography," *Chiang Mai J Sci*, vol. 49, pp. 81-92, 2022.
- [170] R. G. Gary, W. Song, M. J. Giannetto, J. P. Rolland, M. Nedergaard, and K. J. Parker, "Mouse brain elastography changes with sleep/wake cycles, aging, and Alzheimer's disease," *NeuroImage*, vol. 295, p. 120662, 2024.

- [171] G. Schmidt, R. McAuley, B. E. Bouma, and N. Uribe-Patarroyo, "Asynchronous optical coherence elastography and directional phase gradient analysis," (in eng), *J Biomed Opt*, vol. 30, no. 12, p. 124506, Dec 2025, doi: 10.1117/1.Jbo.30.12.124506.
- [172] T. Mekonnen, A. W. Schill, C. Zevallos-Delgado, M. Singh, S. R. Aglyamov, and K. V. Larin, "Reverberant optical coherence elastography using multifocal acoustic radiation force," (in eng), *Opt Lett*, vol. 48, no. 11, pp. 2773-2776, Jun 1 2023, doi: 10.1364/ol.482201.
- [173] T. Mekonnen, C. Zevallos-Delgado, M. Singh, S. R. Aglyamov, and K. V. Larin, "Multifocal acoustic radiation force-based reverberant optical coherence elastography for evaluation of ocular globe biomechanical properties," (in eng), *J Biomed Opt*, vol. 28, no. 9, p. 095001, Sep 2023, doi: 10.1117/1.Jbo.28.9.095001.
- [174] M. Singh, F. Zvietcovich, C. Zevallos-Delgado, Y. S. Ambekar, S. R. Aglyamov, and K. V. Larin, "Whole embryo biomechanics with reverberant optical coherence elastography," *Optica*, vol. 11, no. 5, pp. 686-692, 2024.
- [175] C. D. Caiaffa, G. Tukiman, C. Z. Delgado, Y. S. Ambekar, T. T. Mekonnen, M. Singh, V. Rodriguez, E. Ricco, D. Kraushaar, S. R. Aglyamov, G. Scarcelli, K. V. Larin, R. H. Finnell, and R. M. Cabrera, "Dolutegravir induces FOLR1 expression during brain organoid development," (in eng), *Front Mol Neurosci*, vol. 17, p. 1394058, 2024, doi: 10.3389/fnmol.2024.1394058.

- [176] C. Zevallos-Delgado, T. T. Mekonnen, A. W. Schill, M. Singh, S. R. Aglyamov, and K. V. Larin, "Air-coupled ultrasound based noncontact reverberant optical coherence elastography," *Light: Advanced Manufacturing*, vol. 6, no. 4, pp. 813-820, 2026.
- [177] I. E. Kabir, D. A. Caban-Rivera, J. Ormachea, K. J. Parker, C. L. Johnson, and M. M. Doyley, "Reverberant magnetic resonance elastography," in *Medical Imaging 2023: Physics of Medical Imaging*, 2023, vol. 12463: SPIE, pp. 769-773.
- [178] K. J. Parker, I. E. Kabir, M. M. Doyley, A. Faiyaz, M. N. Uddin, G. Flores, and G. Schifitto, "Brain elastography in aging relates to fluid/solid trendlines," (in eng), *Phys Med Biol*, vol. 69, no. 11, May 29 2024, doi: 10.1088/1361-6560/ad4446.
- [179] H. Asemani, J. P. Rolland, and K. J. Parker, "Integrated Difference Autocorrelation: A Novel Approach to Estimate Shear Wave Speed in the Presence of Compression Waves," (in eng), *IEEE Trans Biomed Eng*, vol. 72, no. 2, pp. 586-594, Feb 2025, doi: 10.1109/tbme.2024.3464104.
- [180] J. Ormachea and F. Zvietcovich, "Reverberant shear wave elastography: a multi-modal and multi-scale approach to measure the viscoelasticity properties of soft tissues," *Frontiers in Physics*, vol. 8, p. 606793, 2021.
- [181] H. Asemani, P. Meemon, G. F. Barrera, J. P. Rolland, and K. J. Parker, "Multi-Frequency Reverberant Shear Waves for Assessing Tissue Dispersion in Optical Coherence Elastography," (in eng), *ArXiv*, Apr 10 2025.

- [182] I. E. Kabir, D. A. Caban-Rivera, J. Ormachea, K. J. Parker, C. L. Johnson, and M. M. Doyley, "Reverberant magnetic resonance elastographic imaging using a single mechanical driver," (in eng), *Phys Med Biol*, vol. 68, no. 5, Feb 27 2023, doi: 10.1088/1361-6560/acbbb7.
- [183] M. Singh, M. S. Hepburn, B. F. Kennedy, and K. V. Larin, "Optical coherence elastography."
- [184] Z.-D. Christian, Z. Fernando, and S. Manmohan, "Assessing porcine iris elasticity and mechanical anisotropy with optical coherence elastography," *Journal of Biomedical Photonics & Engineering*, vol. 7, no. 4, p. 40304, 2021.
- [185] L. Villegas, F. Zvietcovich, S. Marcos, and J. S. Birkenfeld, "Revealing regional variations in scleral shear modulus in a rabbit eye model using multi-directional ultrasound optical coherence elastography," (in eng), *Sci Rep*, vol. 14, no. 1, p. 21010, Sep 9 2024, doi: 10.1038/s41598-024-71343-0.
- [186] A. Nair, F. Zvietcovich, M. Singh, M. P. Weikert, S. R. Aglyamov, and K. V. Larin, "Optical coherence elastography measures the biomechanical properties of the ex vivo porcine cornea after LASIK," (in eng), *J Biomed Opt*, vol. 29, no. 1, p. 016002, Jan 2024, doi: 10.1117/1.Jbo.29.1.016002.
- [187] G. Regnault, R. K. Wang, M. O'Donnell, and I. Pelivanov, "Reconstruction of tensile and shear elastic moduli in anisotropic nearly incompressible media using Rayleigh wave phase and group velocities," (in eng), *J Biomed Opt*, vol. 30, no. 12, p. 124503, Dec 2025, doi: 10.1117/1.Jbo.30.12.124503.

- [188] G. Regnault, M. A. Kirby, R. K. Wang, T. T. Shen, M. O'Donnell, and I. Pelivanov, "Possible depth-resolved reconstruction of shear moduli in the cornea following collagen crosslinking (CXL) with optical coherence tomography and elastography," (in eng), *Biomed Opt Express*, vol. 14, no. 9, pp. 5005-5021, Sep 1 2023, doi: 10.1364/boe.497970.
- [189] M. A. Kirby, I. Pelivanov, G. Regnault, J. J. Pitre, R. T. Wallace, M. O'Donnell, R. K. Wang, and T. T. Shen, "Acoustic Micro-Tapping Optical Coherence Elastography to Quantify Corneal Collagen Cross-Linking: An Ex Vivo Human Study," (in eng), *Ophthalmol Sci*, vol. 3, no. 2, p. 100257, Jun 2023, doi: 10.1016/j.xops.2022.100257.
- [190] M. A. Kirby, G. Regnault, I. Pelivanov, M. O'Donnell, R. K. Wang, and T. T. Shen, "Noncontact Acoustic Micro-Tapping Optical Coherence Elastography for Quantification of Corneal Anisotropic Elasticity: In Vivo Rabbit Study," (in eng), *Transl Vis Sci Technol*, vol. 12, no. 3, p. 15, Mar 1 2023, doi: 10.1167/tvst.12.3.15.
- [191] T. T. Mekonnen, Y. S. Ambekar, C. Zevallos-Delgado, A. Nair, F. Zvietcovich, H. Zarkoob, M. Singh, Y. W. Lim, M. Ferrer, S. R. Aglyamov, G. Scarcelli, M. J. Song, and K. V. Larin, "Dual optical elastography detects TGF- β -induced alterations in the biomechanical properties of skin scaffolds," (in eng), *J Biomed Opt*, vol. 29, no. 9, p. 095002, Sep 2024, doi: 10.1117/1.Jbo.29.9.095002.
- [192] Ł. Ambroziński, I. Pelivanov, S. Song, S. J. Yoon, D. Li, L. Gao, T. T. Shen, R. K. Wang, and M. O'Donnell, "Air-coupled acoustic radiation force for non-

- contact generation of broadband mechanical waves in soft media," (in eng), *Appl Phys Lett*, vol. 109, no. 4, p. 043701, Jul 25 2016, doi: 10.1063/1.4959827.
- [193] F. X. Tian, "A novel method of pipeline obstacle detection using ultrasonic," *Applied Mechanics and Materials*, vol. 373, pp. 819-823, 2013.
- [194] Y. You and J. Riedel, "Approaching phase-imaging through defocusing shadowgraphy for acoustic resonator diagnosis and the capability of direct index-of-refraction measurements," (in eng), *Rev Sci Instrum*, vol. 92, no. 10, p. 103703, Oct 1 2021, doi: 10.1063/5.0058334.
- [195] I. Sanchez, R. Martin, F. Ussa, and I. Fernandez-Bueno, "The parameters of the porcine eyeball," *Graefe's Archive for Clinical and Experimental Ophthalmology*, vol. 249, no. 4, pp. 475-482, 2011.
- [196] R. K. Wang, Z. Ma, and S. J. Kirkpatrick, "Tissue Doppler optical coherence elastography for real time strain rate and strain mapping of soft tissue," *Applied Physics Letters*, vol. 89, no. 14, 2006.
- [197] G. R. Ge, B. Tavakol, D. B. Usher, D. C. Adler, J. P. Rolland, and K. J. Parker, "Assessing corneal cross-linking with reverberant 3D optical coherence elastography," (in eng), *J Biomed Opt*, vol. 27, no. 2, Feb 2022, doi: 10.1117/1.Jbo.27.2.026003.
- [198] G. R. Ge, W. Song, M. J. Giannetto, J. P. Rolland, M. Nedergaard, and K. J. Parker, "Mouse brain elastography changes with sleep/wake cycles, aging, and Alzheimer's disease," (in eng), *Neuroimage*, vol. 295, p. 120662, Jul 15 2024, doi: 10.1016/j.neuroimage.2024.120662.

- [199] Y. Zhu, J. Shi, T. E. G. Alvarez-Arenas, C. Li, H. Wang, H. Cai, D. Zhang, X. He, and X. Wu, "Supershear Rayleigh wave imaging for quantitative assessment of biomechanical properties of brain using air-coupled optical coherence elastography," (in eng), *APL Bioeng*, vol. 7, no. 4, p. 046107, Dec 2023, doi: 10.1063/5.0160213.
- [200] M. A. Kirby, K. Zhou, J. J. Pitre, L. Gao, D. Li, I. Pelivanov, S. Song, C. Li, Z. Huang, T. Shen, R. Wang, and M. O'Donnell, "Spatial resolution in dynamic optical coherence elastography," (in eng), *J Biomed Opt*, vol. 24, no. 9, pp. 1-16, Sep 2019, doi: 10.1117/1.Jbo.24.9.096006.
- [201] G. Regnault, M. A. Kirby, M. Kuriakose, T. Shen, R. K. Wang, M. O'Donnell, and I. Pelivanov, "Spatial resolution in optical coherence elastography of bounded media," (in eng), *Biomed Opt Express*, vol. 13, no. 9, pp. 4851-4869, Sep 1 2022, doi: 10.1364/boe.469019.
- [202] X. Feng, G. Y. Li, and S. H. Yun, "Ultra-wideband optical coherence elastography from acoustic to ultrasonic frequencies," (in eng), *Nat Commun*, vol. 14, no. 1, p. 4949, Aug 16 2023, doi: 10.1038/s41467-023-40625-y.
- [203] N. Learthrapun and S. G. Adie, "Recent advances in optical elastography and emerging opportunities in the basic sciences and translational medicine [Invited]," (in eng), *Biomed Opt Express*, vol. 14, no. 1, pp. 208-248, Jan 1 2023, doi: 10.1364/boe.468932.

- [204] B. F. Kennedy, P. Wijesinghe, and D. D. Sampson, "The emergence of optical elastography in biomedicine," *Nature Photonics*, vol. 11, no. 4, pp. 215-221, 2017.
 - [205] N. Nitta, M. Yamakawa, H. Hachiya, and T. Shiina, "A review of physical and engineering factors potentially affecting shear wave elastography," (in eng), *J Med Ultrason (2001)*, vol. 48, no. 4, pp. 403-414, Oct 2021, doi: 10.1007/s10396-021-01127-w.
 - [206] A. Curatolo, M. Villiger, D. Lorensen, P. Wijesinghe, A. Fritz, B. F. Kennedy, and D. D. Sampson, "Ultrahigh-resolution optical coherence elastography," (in eng), *Opt Lett*, vol. 41, no. 1, pp. 21-4, Jan 1 2016, doi: 10.1364/ol.41.000021.
 - [207] M. F. Donovan and M. Cascella, "Embryology, Weeks 6-8," in *StatPearls*. Treasure Island (FL) ineligible companies. Disclosure: Marco Cascella declares no relevant financial relationships with ineligible companies.: StatPearls Publishing
- Copyright © 2026, StatPearls Publishing LLC., 2026.
- [208] N. Gjorevski and C. M. Nelson, "The mechanics of development: Models and methods for tissue morphogenesis," (in eng), *Birth Defects Res C Embryo Today*, vol. 90, no. 3, pp. 193-202, Sep 2010, doi: 10.1002/bdrc.20185.
 - [209] T. Yamane, "Mouse Yolk Sac Hematopoiesis," (in eng), *Front Cell Dev Biol*, vol. 6, p. 80, 2018, doi: 10.3389/fcell.2018.00080.

- [210] V. Wittamer and J. Y. Bertrand, "Yolk sac hematopoiesis: does it contribute to the adult hematopoietic system?," (in eng), *Cell Mol Life Sci*, vol. 77, no. 20, pp. 4081-4091, Oct 2020, doi: 10.1007/s00018-020-03527-6.
- [211] M. C. Yoder, K. Hiatt, and P. Mukherjee, "In vivo repopulating hematopoietic stem cells are present in the murine yolk sac at day 9.0 postcoitus," (in eng), *Proc Natl Acad Sci U S A*, vol. 94, no. 13, pp. 6776-80, Jun 24 1997, doi: 10.1073/pnas.94.13.6776.
- [212] A. Ornoy and R. K. Miller, "Yolk sac development, function and role in rodent pregnancy," (in eng), *Birth Defects Res*, vol. 115, no. 14, pp. 1243-1254, Aug 15 2023, doi: 10.1002/bdr2.2172.
- [213] M. C. Wallingford and C. M. Giachelli, "Loss of PiT-1 results in abnormal endocytosis in the yolk sac visceral endoderm," (in eng), *Mech Dev*, vol. 133, pp. 189-202, Aug 2014, doi: 10.1016/j.mod.2014.08.001.
- [214] B. Suguna and K. Sukanya, "Yolk sac size & shape as predictors of first trimester pregnancy outcome: A prospective observational study," (in eng), *J Gynecol Obstet Hum Reprod*, vol. 48, no. 3, pp. 159-164, Mar 2019, doi: 10.1016/j.jogoh.2018.10.016.
- [215] J. Sutlive, H. Xiu, Y. Chen, K. Gou, F. Xiong, M. Guo, and Z. Chen, "Generation, Transmission, and Regulation of Mechanical Forces in Embryonic Morphogenesis," (in eng), *Small*, vol. 18, no. 6, p. e2103466, Feb 2022, doi: 10.1002/sml.202103466.

- [216] M. Li, C. Zhang, L. Wang, L. Liu, N. Xi, Y. Wang, and Z. Dong, "Investigating the morphology and mechanical properties of blastomeres with atomic force microscopy," *Surface and interface analysis*, vol. 45, no. 8, pp. 1193-1196, 2013.
- [217] Y. Tan, C. W. Kong, S. Chen, S. H. Cheng, R. A. Li, and D. Sun, "Probing the mechanobiological properties of human embryonic stem cells in cardiac differentiation by optical tweezers," (in eng), *J Biomech*, vol. 45, no. 1, pp. 123-8, Jan 3 2012, doi: 10.1016/j.jbiomech.2011.09.007.
- [218] V. S. Chen, J. P. Morrison, M. F. Southwell, J. F. Foley, B. Bolon, and S. A. Elmore, "Histology Atlas of the Developing Prenatal and Postnatal Mouse Central Nervous System, with Emphasis on Prenatal Days E7.5 to E18.5," (in eng), *Toxicol Pathol*, vol. 45, no. 6, pp. 705-744, Aug 2017, doi: 10.1177/0192623317728134.
- [219] S. Tani, U. I. Chung, S. Ohba, and H. Hojo, "Understanding paraxial mesoderm development and sclerotome specification for skeletal repair," (in eng), *Exp Mol Med*, vol. 52, no. 8, pp. 1166-1177, Aug 2020, doi: 10.1038/s12276-020-0482-1.
- [220] S. Siddiqui, H. Siddiqui, E. Riguene, and M. Nomikos, "Zebrafish: A Versatile and Powerful Model for Biomedical Research," (in eng), *Bioessays*, vol. 47, no. 12, p. e70080, Dec 2025, doi: 10.1002/bies.70080.
- [221] K. A. McGowan and G. S. Barsh, "Evolution: How the zebrafish got its stripes," (in eng), *Elife*, vol. 5, Feb 15 2016, doi: 10.7554/eLife.14239.

- [222] K. Li, B. Liu, Z. Wang, Y. Li, H. Li, S. Wu, and Z. Li, "Quantitative characterization of zebrafish development based on multiple classifications using Mueller matrix OCT," (in eng), *Biomed Opt Express*, vol. 14, no. 6, pp. 2889-2904, Jun 1 2023, doi: 10.1364/boe.488614.
- [223] L. Kagemann, H. Ishikawa, J. Zou, P. Charukamnoetkanok, G. Wollstein, K. A. Townsend, M. L. Gabriele, N. Bahary, X. Wei, J. G. Fujimoto, and J. S. Schuman, "Repeated, noninvasive, high resolution spectral domain optical coherence tomography imaging of zebrafish embryos," (in eng), *Mol Vis*, vol. 14, pp. 2157-70, 2008.
- [224] Y. Tomizawa, D. F. Daggett, G. Zheng, and K. Hoshino, "Light microscopy-based elastography for the mechanical characterization of zebrafish somitogenesis," (in eng), *J Biophotonics*, vol. 16, no. 3, p. e202200238, Mar 2023, doi: 10.1002/jbio.202200238.
- [225] M. Li, L. Zhao, P. S. Page-McCaw, and W. Chen, "Zebrafish Genome Engineering Using the CRISPR-Cas9 System," (in eng), *Trends Genet*, vol. 32, no. 12, pp. 815-827, Dec 2016, doi: 10.1016/j.tig.2016.10.005.
- [226] S. Liu, R. Narumi, N. Ikeda, O. Morita, and J. Tasaki, "Chemical-induced craniofacial anomalies caused by disruption of neural crest cell development in a zebrafish model," (in eng), *Dev Dyn*, vol. 249, no. 7, pp. 794-815, Jul 2020, doi: 10.1002/dvdy.179.

- [227] V. Rajesh and P. K. Divya, "Embryonic exposure to decitabine induces multiple neural tube defects in developing zebrafish," (in eng), *Fish Physiol Biochem*, vol. 49, no. 6, pp. 1357-1379, Dec 2023, doi: 10.1007/s10695-023-01261-x.
- [228] J. H. Yin and K. A. Horzmann, "Embryonic Zebrafish as a Model for Investigating the Interaction between Environmental Pollutants and Neurodegenerative Disorders," (in eng), *Biomedicines*, vol. 12, no. 7, Jul 13 2024, doi: 10.3390/biomedicines12071559.
- [229] R. Kopp, J. Legler, and J. Legradi, "Alterations in locomotor activity of feeding zebrafish larvae as a consequence of exposure to different environmental factors," (in eng), *Environ Sci Pollut Res Int*, vol. 25, no. 5, pp. 4085-4093, Feb 2018, doi: 10.1007/s11356-016-6704-3.
- [230] D. S. Kleinhans and V. Lecaudey, "Standardized mounting method of (zebrafish) embryos using a 3D-printed stamp for high-content, semi-automated confocal imaging," (in eng), *BMC Biotechnol*, vol. 19, no. 1, p. 68, Oct 22 2019, doi: 10.1186/s12896-019-0558-y.
- [231] J. Jutoy, H. Mehrabi, and E. Jung, "Agarose Stamped Device: A simple but effective fixation method for Zebrafish Larvae through customizable 3D printed molds for *Danio rerio* (Zebrafish)," *bioRxiv*, p. 2025.03. 04.641502, 2025.